A STEP-BY-STEP GUIDE

Excel Full Course Notes

From the interface to dashboards, one clear page at a time

1. What Are These Notes?

These notes walk through Excel from the interface to dashboards. This volume holds Modules 1 to 4 (pages 3 to 14). Every page follows the same pattern: what the topic is, the syntax or key shortcuts, worked examples, when to use it, important points, common mistakes and a Data Analyst tip.

2. Contents

No.	Module	Page
1	Excel Basics & Interface	3
2	Entering & Formatting Data	6
3	Formulas Fundamentals	9
4	Essential Functions	12
5	Logical Functions	15
6	Conditional Math Functions	18
7	Lookup & Reference	20
8	Text Functions	23
9	Date & Time Functions	26
10	Data Management Tools	29
11	Dynamic Array Functions	33
12	Pivot Tables	36
13	Charts & Visualisation	38
14	Power Query & Power Pivot	40
15	What-If Analysis & Macros	42
16	Building a Dashboard	44
17	Shortcuts, Practice & Interview Prep	46

3. How to Read These Notes

Style	Meaning
Bold text	A key term to remember
`=SUM(A1:A5)`	A formula or key you can type
Important Things to Remember	The points most likely to matter in practice and in exams
Common Mistakes	Errors learners often make, with the fix
Data Analyst Tip	A habit that makes your analysis more reliable

Excel Basics

Module 1 | Know your way around before you calculate

1. What is Excel?

Excel is a **spreadsheet** program from Microsoft. It is used to store, calculate, analyse and visualise data. Everything lives in a grid made of **rows** and **columns**. Rows run across the sheet and are numbered (1, 2, 3...). Columns run down the sheet and are lettered (A, B, C...).

2. Key Terms

Term	Meaning	Example
Workbook	The whole Excel file	`Sales.xlsx`
Worksheet	One sheet (tab) inside a workbook	`Sheet1`
Cell	One box where a row and a column meet	`B3`
Range	A group of cells (the colon means "to")	`A1:C10`
Active cell	The cell with the thick green border, where you type	`Where you type`

3. Simple Example

	A	B
1	Item	Qty
2	Pen	10
3	Book	5
4	**Total**	**15**

Formula: `=SUM(B2:B3)`
B2 is a **cell**. B2:B3 is a **range** (B2 "to" B3). Excel adds 10 + 5.
Result: 15

4. Size Limits (Good to Know)

Item	Limit
Rows per sheet	1,048,576
Columns per sheet	16,384 (A to XFD)
Characters in one cell	32,767
Sheets in a workbook	Limited only by memory

5. When to Use It

- Store lists such as customers, orders or marks in rows and columns
- Calculate totals, averages and percentages automatically
- Analyse data with sorting, filtering, PivotTables and charts
- Share results as tables, charts or dashboards

6. Important Things to Remember

- A cell address is the column letter first, then the row number: B3, not 3B.
- A range uses a colon: `A1:C10` means every cell from A1 to C10.
- The active cell is where anything you type will go.
- A worksheet is one tab. A workbook is the whole file with all its tabs.

7. Common Mistakes

Mistake: Writing the row number before the column letter, such as 3B. **Fix:** Always write the letter first: B3.

Mistake: Typing into the wrong cell because you did not notice which cell was active. **Fix:** Look for the thick green border, or check the Name Box before typing.

8. Data Analyst Tip

Know the limits before you import data. A sheet holds 1,048,576 rows, and a larger CSV file will be cut off when opened in Excel. For bigger data, load it with Power Query (Module 14).

Interface Tour & File Types

Module 1 | Know your way around before you calculate

1. What is the Interface?

The interface is everything you see around the grid: the toolbars, the Ribbon, the bars that show cell contents and the tabs at the bottom. Knowing where each part is saves a lot of time.

2. Parts of the Screen

Part	What it does
Quick Access Toolbar	Save, Undo and Redo at the very top
Ribbon	Tabs that hold all commands: Home, Insert, Page Layout, Formulas, Data, Review, View
Name Box	Shows the active cell address. Type an address here to jump there
Formula Bar	Shows what is really inside the active cell
Sheet tabs	Bottom left. Click + to add a new sheet
Status Bar	Bottom. Shows Sum, Average and Count of the selected cells

3. What Each Ribbon Tab is For

Tab	Use it for
Home	Fonts, alignment, number format, styles, sort and filter
Insert	Tables, PivotTables, charts, sparklines, shapes
Page Layout	Margins, orientation, print area, themes
Formulas	Function library, Name Manager, auditing tools
Data	Import data, sort, filter, validation, what-if
Review	Spell check, notes, protect sheet
View	Freeze panes, zoom, gridlines, page break preview

4. File Types

Type	What it is
`.xlsx`	Normal workbook
`.xlsm`	Workbook with macros
`.csv`	Plain text, values separated by commas, no formatting
`.xls`	Old Excel 97-2003 format
`.pdf`	Export to share a read-only copy

5. Simple Example: Status Bar

	A	B
1	Student	Marks
2	Rahim	78
3	Karim	85
4	Hasan	92
5	Nabil	66
6	Sara	79

Select B2:B6 and look at the Status Bar. It shows **Average: 80, Count: 5, Sum: 400**. No formula is needed.

6. When to Use It

- Check a total or average quickly without typing a formula
- Jump to a far-away cell using the Name Box
- Find the right command by knowing its Ribbon tab
- Choose the correct file type before saving or sharing

7. Important Things to Remember

- The cell shows a result, but the Formula Bar shows what is really stored.
- Save as .xlsm only when the workbook contains macros. A normal .xlsx cannot hold them.
- A .csv file keeps values only. Formulas, colours and extra sheets are lost.

8. Common Mistakes

Mistake: Saving a workbook with formatting and several sheets as .csv. **Fix:** Save as .xlsx and export a .csv copy only when another program needs it.

9. Data Analyst Tip

Select a column of numbers and check the Status Bar. If Count is lower than the number of rows, some cells are blank or stored as text, so investigate before you analyse.

Navigation Shortcuts

Module 1 | Know your way around before you calculate

1. What are Navigation Shortcuts?

Navigation shortcuts move or select around a sheet using only the keyboard. In a sheet with thousands of rows they are far faster than scrolling with the mouse.

2. Shortcuts

Shortcut	What it does
`Ctrl + Arrow key`	Jump to the edge of the data block
`Ctrl + Home`	Go to cell A1
`Ctrl + End`	Go to the last used cell
`Ctrl + Space`	Select the whole column
`Shift + Space`	Select the whole row
`Ctrl + A`	Select all (press twice inside a table)
`Ctrl + PgDn / PgUp`	Next / previous sheet
`F5 or Ctrl + G`	Go To dialog

3. Simple Example

	A	B
1	Name	Marks
2	Rahim	70
3	Karim	80
4	Hasan	60
5		
6		
7	Sara	90

Select B2 and press `Ctrl + Down Arrow`. The cursor jumps to B4, the end of the block. Press it again and it jumps to B7, the next filled cell.

4. Another Example

Press `F5` (or `Ctrl + G`), type `D500` and press Enter. Excel jumps straight to cell D500.

Press `Ctrl + Home` to return to A1, and `Ctrl + End` to see the last cell Excel considers used.

5. When to Use It

- Reach the bottom of a long list in one keystroke
- Select an entire column or row before formatting it
- Move between sheets without the mouse
- Check whether stray cells exist far from your data

6. Important Things to Remember

- `Ctrl + Arrow` stops at blank cells inside the data. Press it again to jump to the next filled cell.
- `Ctrl + End` goes to the last **used** cell. This can be further than your visible data if old formatting is left behind.
- Inside a table, `Ctrl + A` first selects the table data. Press it a second time to select the whole sheet.
- Add `Shift` to any `Ctrl + Arrow` move to select cells along the way.

7. Common Mistakes

Mistake: Pressing Ctrl + Down Arrow and assuming the end of the data was reached, when a blank cell stopped the jump. **Fix:** Check for blank rows, or press Ctrl + End to confirm the real last cell.

Mistake: Scrolling with the mouse through thousands of rows. **Fix:** Use Ctrl + Arrow, Ctrl + Home and Ctrl + End.

8. Data Analyst Tip

Select a whole data column quickly with `Ctrl + Shift + Down Arrow` from its first cell. You then see the exact range to use in a formula, and any blank cell that breaks the selection stands out immediately.

Entering Data & AutoFill

Module 2 | Type fast, format clean

1. What is Data Entry?

Data entry is how values get into cells. Excel decides how to treat what you type, so it helps to know the types of data and the keys that make typing faster.

2. Types of Data

Type	Examples	Notes
Text	Names, codes	Left aligned by default
Numbers	1250, 3.5	Right aligned by default
Dates and times	01-01-2026	Stored as numbers, shown as dates
Logical	TRUE, FALSE	Results of comparisons
Errors	#N/A, #DIV/0!	Shown when a formula cannot work

3. Entering and Editing

Key	What it does
`Enter / Tab / Esc`	Enter moves down, Tab moves right, Esc cancels typing
`F2`	Edits the cell without double clicking
`Alt + Enter`	Starts a new line inside the same cell
`Ctrl + ;`	Enters today's date
`Ctrl + Shift + ;`	Enters the current time
`'007`	Starts with an apostrophe to keep leading zeros as text
`Ctrl + D / Ctrl + R`	Fill the cell above downwards / fill from the left

4. Simple Example: AutoFill

	A
1	Value
2	10
3	20
4	30
5	40

Type 10 and 20 first so Excel learns the pattern. Select both and drag the **fill handle** (the small square at the corner of the active cell). Excel continues 30, 40 and so on. It also continues 1, 2, 3 or Jan, Feb, Mar or Mon, Tue, Wed.

5. Another Example: Flash Fill

	A	B
1	Full Name	First Name
2	Rahul Sharma	Rahul
3	Priya Verma	Priya
4	Amit Das	Amit

Type `Rahul` next to `Rahul Sharma`. In the next row press `Ctrl + E`. **Flash Fill** copies the pattern down the column.

6. When to Use It

- Number rows, dates or months without typing each one
- Split or join text such as first and last names with Flash Fill
- Keep codes like 007 exactly as typed
- Stamp the current date or time into a sheet

7. Important Things to Remember

- Dates and times are numbers underneath. Only the format makes them look like dates.
- Starting with an apostrophe makes the entry text, so leading zeros are kept.
- Flash Fill works only when Excel can see a clear pattern in your first example.

8. Common Mistakes

Mistake: Typing 007 and finding Excel shows 7. **Fix:** Type '007 (apostrophe first) or format the cell as Text before typing.

Mistake: Dragging the fill handle on a single number and getting copies instead of a series. **Fix:** Type two values first, select both, then drag.

9. Data Analyst Tip

Use `Ctrl + ;` for a date that stays fixed. The `TODAY()` function changes every day, which can silently change old reports.

Number Formats & Layout

Module 2 | Type fast, format clean

1. What is Number Formatting?

Number formatting changes how a value **looks** without changing the value itself. Press `Ctrl + 1` to open Format Cells. The real value stays the same inside the cell, so always check the Formula Bar.

2. Number Formats

Format	Typed 1234.5 shows as
General	1234.5
Number (2 decimals)	1234.50
Currency	$1,234.50
Percentage (typed 0.25)	25%
Date (typed 46023)	01-01-2026
Custom `#,##0`	1,235
Custom `0.00 "kg"`	1234.50 kg

3. Simple Example

	A	B
1	Typed	Shows as
2	0.25	25%
3	1234.5	1234.50 kg
4	**=A3*2**	**2469**

Formula: `=A3*2`
Cell A3 shows `1234.50 kg` but still holds 1234.5. Excel uses the real value. **Result: 2469**

4. Alignment and Layout

Feature	What it does
Wrap Text	Keeps long text inside the cell by growing the row height
Center Across Selection	Centres text over several cells without merging
AutoFit	Double click a column border to fit the width

5. Insert, Delete and Hide Shortcuts

Shortcut	What it does
`Ctrl + Shift + +`	Inserts cells
`Ctrl + -`	Deletes cells
`Ctrl + 9`	Hides a row
`Ctrl + 0`	Hides a column

6. When to Use It

- Show currency, percentages or units without changing the numbers
- Make long headings readable with Wrap Text
- Tidy a sheet by hiding helper rows and columns
- Control how many decimals appear in reports

7. Important Things to Remember

- Formatting only changes how a value looks. The real value is shown in the Formula Bar.
- **Merge & Center** looks neat but breaks sorting and filtering. Prefer **Center Across Selection** (Format Cells > Alignment).
- Typing 0.25 in a Percentage cell shows 25%.

8. Common Mistakes

Mistake: Assuming a number is rounded because it shows 2 decimals. **Fix:** Use `=ROUND(A1,2)` when the stored value must change.

Mistake: Merging header cells and then being unable to sort the table. **Fix:** Use Center Across Selection instead.

9. Data Analyst Tip

If a total looks off by a small amount, click the cells and compare what the Formula Bar holds with what the cell shows. Hidden decimals are a common cause.

Paste Special & Find and Replace

Module 2 | Type fast, format clean

1. What are They?

Paste Special (`Ctrl + Alt + V`) lets you paste only part of what you copied, such as values or formats. **Find and Replace** (`Ctrl + F` and `Ctrl + H`) searches your data and changes it in bulk.

2. Paste Special Options

Option	Why use it
Values	Keep the result, drop the formula
Formulas	Copy formulas without formatting
Formats	Copy only the look
Column widths	Make widths match
Transpose	Swap rows and columns
Multiply / Add	Change a whole range by a number in one go

3. Simple Example: Multiply

	A	B
1	Price	After x 1.18
2	100	118
3	200	236
4	300	354

Copy a cell holding `1.18`. Select the prices, choose Paste Special > **Multiply**. Each price is multiplied by 1.18 (18% tax) in one step.

4. Wildcards in Find

Wildcard	Meaning	Finds
*	Any characters	`Sh*` finds Sharma, Shah
?	One character	`R?hul` finds Rahul
~*	A real star	`~*` finds the symbol *

5. Find and Replace Basics

- `Ctrl + F` finds and `Ctrl + H` replaces.
- Click **Options** to search the whole workbook or inside formulas.
- To find the wildcard characters themselves, put a tilde before them: `~*` finds a real star.

6. When to Use It

- Convert formulas to fixed numbers with Paste Special > Values
- Flip a table from rows to columns with Transpose
- Apply a price rise or tax to a whole range at once
- Clean data by replacing a wrong word or symbol everywhere

7. Important Things to Remember

- Values removes the formula, so the number will no longer update.
- Wildcards: * means any characters and ? means one character.
- Find and Replace searches only the current sheet unless you change Within to Workbook.

8. Common Mistakes

Mistake: Clicking Replace All without checking what will change. **Fix:** Use Find All first to review the matches, or replace one at a time.

Mistake: Copying formulas to another file and getting broken references. **Fix:** Paste as Values when the numbers only need to be kept.

9. Data Analyst Tip

Keep a backup copy of the sheet before a large Replace All or Multiply paste. Both are easy to run and hard to undo after you save and close.

Formula Basics & Operators

Module 3 | Operators, references and errors

1. What is a Formula?

A formula is an instruction that tells Excel to calculate something. **Every formula starts with an equals sign.** Without it, Excel treats your entry as plain text.

2. Syntax

=A2+B2

This adds the values in A2 and B2. If A2 changes, the answer updates automatically.

Operator	Meaning	Example
+ -	Add, subtract	`=10-4 gives 6`
* /	Multiply, divide	`=6*3 gives 18`
^	Power	`=2^3 gives 8`
&	Join text	`="Hi "&A1`
%	Percent	`=50%*200 gives 100`
= > < >= <= <>	Comparisons	`=A1>=40 gives TRUE`

3. Simple Example

	A	B	C
1	Price	Tax	Total
2	**100**	**18**	**118**

Formula in C2: `=A2+B2`
Excel adds 100 + 18. **Result: 118**. Change A2 to 200 and C2 updates to 218.

4. Order of Operations

Excel follows **BODMAS**: Brackets, then Powers, then Multiply / Divide (left to right), then Add / Subtract.

Formula	Result	Why
`=2+3*4`	14	3*4 happens first
`=(2+3)*4`	20	Brackets go first

5. When to Use It

- Add, subtract, multiply or divide values that may change later
- Join text, such as a greeting and a name
- Compare values to get TRUE or FALSE
- Build totals, taxes and discounts that update automatically

6. Important Things to Remember

- Forgetting the equals sign makes Excel show your formula as text.
- Use brackets to control the order. `=(2+3)*4` gives 20, but `=2+3*4` gives 14.
- Text inside a formula needs double quotes, such as `="Hi "&A1`.
- A comparison such as `=A1>=40` returns TRUE or FALSE, not a number.

7. Common Mistakes

Mistake: Typing 2+3*4 without the equals sign and seeing the text instead of 14. **Fix:** Start every formula with =.

Mistake: Expecting (2+3)*4 when typing =2+3*4. **Fix:** Add brackets around the part that must be calculated first.

8. Data Analyst Tip

Never type numbers that already sit in cells into a formula. Point to the cell, as in `=A2+B2`, so the result updates when the data changes.

Cell References & Named Ranges

Module 3 | Operators, references and errors

1. What are Cell References?

A cell reference tells a formula where to look. What happens to a reference when you copy the formula depends on its type.

Type	Looks like	When you copy it
Relative	`A1`	Changes with position
Absolute	`$A$1`	Never changes
Mixed (column fixed)	`$A1`	Row changes only
Mixed (row fixed)	`A$1`	Column changes only

2. Syntax

Press `F4` while editing a reference to cycle: `A1` > `$A$1` > `A$1` > `$A1`.

3. Simple Example: Absolute

	A	B	C
1	Item	Price	Tax
2	Pen	100	18
3	Book	200	36
4	**Bag**	**300**	**54**

Tax rate 18% is in E1. Formula in C2: `=B2*$E$1`
The $ locks E1, so copying down still points to the same rate.

4. Another Example: Relative

Copy `=A2*B2` from C2 to C3 and Excel writes `=A3*B3` for you. That is a **relative reference** at work.

5. Other Sheets and Files

Where	Example
Another sheet	`=Sheet2!A1`
Name with spaces	`='Jan Sales'!A1`
Another workbook	`='[Budget.xlsx]Sheet1'!A1`

6. Named Ranges

- Select the cells, type a name in the Name Box and press Enter.
- Then use it: `=SUM(Sales)` instead of `=SUM(C2:C100)`.
- `Ctrl + F3` opens Name Manager to edit or delete names.
- Names cannot have spaces and cannot look like a cell address.

7. When to Use It

- Apply one tax rate, discount or target to a whole column
- Pull numbers from other sheets or workbooks
- Make long formulas readable with named ranges

8. Important Things to Remember

- Relative (A1) changes when copied, absolute ($A$1) never changes.
- Mixed references lock only the column ($A1) or only the row (A$1).
- Copying a formula only adjusts relative parts of its references.

9. Common Mistakes

Mistake: Copying `=B2*E1` down so the rate moves to E2, E3 and returns 0 or wrong values. **Fix:** Lock the rate with `=B2*$E$1` (press F4).

Mistake: Naming a range with a space or a name like AB12. **Fix:** Use a name such as `Sales_Q1`.

10. Data Analyst Tip

Put assumptions such as tax rate or target in their own labelled cells and point to them with absolute references. You can then change one cell and update the whole model.

Common Errors & Auditing Tools

Module 3 | Operators, references and errors

1. What are Formula Errors?

When Excel cannot complete a calculation it shows an error code instead of a result. Each code points to a different cause, so reading it tells you where to look.

2. Common Errors and Fixes

Error	Meaning	Usual fix
`#####`	Column too narrow	Widen the column
`#DIV/0!`	Divided by zero or blank	Check divisor, or use IFERROR
`#N/A`	Value not found	Check lookup value and spelling
`#NAME?`	Unknown function or name	Fix the typo
`#REF!`	Reference was deleted	Rebuild the reference
`#VALUE!`	Wrong data type	Numbers stored as text?
`#NUM!`	Invalid number	Check the arguments
`#SPILL!`	Spill range is blocked	Clear the cells in the way

3. Simple Example: #DIV/0!

	A	B	C
1	Sales	Orders	Average
2	**500**	**0**	**#DIV/0!**

Formula in C2: `=A2/B2`
Dividing by 0 gives an error. Fix: `=IFERROR(A2/B2,0)` returns **0**.

4. Another Example: #VALUE!

	A	B	C
1	Qty	Note	Total
2	**10**	**ten**	**#VALUE!**

Formula in C2: `=A2+B2`
B2 holds text, so Excel cannot add it. Replace the text with a number.

5. Auditing Tools (Formulas Tab)

Tool	What it does
Trace Precedents / Dependents	Draw arrows to related cells
Evaluate Formula	Walks through a formula step by step
Ctrl + ` (key under Esc)	Shows all formulas on the sheet
F9 on part of a formula	Shows that part's value. Press Esc afterwards, do not press Enter
Calculation Options	If numbers do not update, it may be set to Manual. Press F9 to recalculate

6. When to Use It

- Find out why a formula returns an error
- Check which cells feed a result and which cells depend on it
- Review a whole sheet of formulas at once

7. Important Things to Remember

- `#####` is not a real error. The column is simply too narrow to show the number.
- Pressing Enter after F9 replaces that part of the formula with its value. Press Esc instead.
- If numbers do not update, check Formulas > Calculation Options.

8. Common Mistakes

Mistake: Wrapping every formula in IFERROR so errors disappear. **Fix:** Find the cause first. Use IFERROR only where an error is expected.

Mistake: Pressing Enter after F9 and overwriting part of the formula. **Fix:** Press Esc to cancel.

9. Data Analyst Tip

Read the error code before you change anything. The code tells you the kind of problem, and Evaluate Formula then shows the exact step where it goes wrong.

Functions: Anatomy & Statistics

Module 4 | The everyday toolkit

1. What is a Function?

A function is a ready-made formula with a name. You give it values to work on, and it returns a result. It saves you from writing long calculations by hand.

2. Syntax

```
=FUNCTION(argument1, argument2, ...)
```

Name first, then brackets. Arguments are separated by commas. Type `=` and the first letters, then press `Tab` to accept the suggestion. `Shift + F3` opens the Insert Function box. A colon means a range (`A1:A5`). A comma means separate cells (`A1,A5`).

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	78
3	Karim	85
4	Hasan	92
5	Nabil	66
6	Sara	79

Marks are in B2:B6: 78, 85, 92, 66, 79. Every formula below uses this range.

4. What Each Formula Returns

Formula	Result	What it does
`=SUM(B2:B6)`	400	Adds all numbers
`=AVERAGE(B2:B6)`	80	Mean value
`=MAX(B2:B6)`	92	Largest
`=MIN(B2:B6)`	66	Smallest
`=MEDIAN(B2:B6)`	79	Middle value
`=LARGE(B2:B6,2)`	85	2nd largest
`=SMALL(B2:B6,2)`	78	2nd smallest
`=RANK.EQ(85,B2:B6)`	2	Rank, largest = 1

5. When to Use It

- Total sales, marks or expenses with SUM
- Find the average, highest and lowest values in a list
- Find the top or bottom 2nd, 3rd... values with LARGE and SMALL
- Rank students, products or salespeople

6. Important Things to Remember

- AVERAGE is the SUM divided by how many numbers there are: 400 ÷ 5 = 80.
- MEDIAN is the middle value once the numbers are in order: 66, 78, 79, 85, 92.
- RANK.EQ gives rank 1 to the largest value by default.
- Every opening bracket needs a closing bracket.

7. Common Mistakes

Mistake: Using a comma when you meant a range, such as `=SUM(B2,B6)` instead of `=SUM(B2:B6)`. **Fix:** Use a colon for a continuous range and commas for separate cells.

Mistake: Including the header row or a Total row in the range. **Fix:** Select only the data rows.

8. Data Analyst Tip

Compare the AVERAGE with the MEDIAN. Here they are close (80 and 79), so the data is fairly even. A big gap means a few extreme values are pulling the average.

Counting, Rounding & SUBTOTAL

Module 4 | The everyday toolkit

1. What are These Functions?

These helpers count cells, tidy numbers and total only the rows you can see. They are used in almost every report.

2. Counting Functions

Function	Counts
`COUNT`	Cells that contain numbers
`COUNTA`	Cells that are not empty
`COUNTBLANK`	Empty cells
`ROWS / COLUMNS`	Number of rows / columns in a range

3. Simple Example: Counting

	A	B
1	Name	Marks
2	Rahim	70
3	Karim	80
4	Hasan	
5	Nabil	Absent
6	Sara	90

Results for B2:B6

Formula	Result
`=COUNT(B2:B6)`	3
`=COUNTA(B2:B6)`	4
`=COUNTBLANK(B2:B6)`	1

COUNT sees only the numbers 70, 80 and 90. COUNTA also counts the text Absent. COUNTBLANK counts the empty B4.

4. Rounding and Number Helpers

Formula	Result	Note
`=ROUND(12.456,2)`	12.46	Normal rounding
`=ROUNDUP(12.451,2)`	12.46	Always away from zero
`=ROUNDDOWN(12.459,2)`	12.45	Always toward zero
`=ROUND(1250,-2)`	1300	Negative = left of the point
`=INT(-7.2)`	-8	Rounds down to an integer
`=MOD(10,3)`	1	Remainder after dividing
`=ABS(-5)`	5	Removes the minus sign

5. SUBTOTAL: Totals That Respect Filters

`=SUBTOTAL(9, B2:B100)`

Code 9 = SUM, 1 = AVERAGE, 2 = COUNT, 3 = COUNTA, 4 = MAX, 5 = MIN. It only counts rows left visible after a filter. Use codes 101-111 to also ignore rows you hid by hand. For example, 109 is SUM that skips hidden rows.

6. When to Use It

- Count filled, empty or numeric cells in a column
- Round prices or percentages for reports
- Total only the filtered rows
- Test for even or odd numbers: `=MOD(A1,2)=0` is TRUE for even numbers

7. Important Things to Remember

- COUNT counts numbers only, COUNTA counts anything that is not empty.
- A negative digit in ROUND works left of the decimal point: `=ROUND(1250,-2)` gives 1300.
- SUM adds hidden and filtered rows too. SUBTOTAL skips filtered rows.

8. Common Mistakes

Mistake: Using SUM on a filtered list and getting the total of all rows. **Fix:** Use `=SUBTOTAL(9, range)`.

Mistake: Formatting to 2 decimals and expecting the stored number to be rounded. **Fix:** Use ROUND to change the real value.

9. Data Analyst Tip

Place `=COUNTA(B2:B6)` next to `=COUNT(B2:B6)`. If COUNT is smaller than COUNTA, some numbers are stored as text.

Percentages, PMT & Random Numbers

Module 4 | The everyday toolkit

1. What are They?

Percentage formulas answer share, growth, tax and discount questions. **PMT** calculates a regular loan payment. **RAND** and **RANDBETWEEN** create random numbers for tests and samples.

2. Syntax

```
=PMT(rate/12, years*12, -loan)
```

This gives the monthly payment. Keep the units matching: if payments are monthly, divide the yearly rate by 12 and multiply the years by 12.

Task	Formula	Example
Share of total	`=B2/SUM($B$2:$B$6)`	Format as %. 78 of 400 = 19.5%
Growth	`=New/Old-1`	From 120 to 150 gives 25%
Add 18% tax	`=B2*1.18`	100 becomes 118
Discount 10%	`=B2*(1-10%)`	100 becomes 90

3. Simple Example: Growth

	A	B	C
1	Old	New	Growth
2	**120**	**150**	**25%**

Formula in C2: `=B2/A2-1`
150 ÷ 120 = 1.25, minus 1 = 0.25. **Result: 25%**

4. Another Example: PMT

	A	B	C	D
1	Loan	Rate	Years	Payment
2	**500000**	**10%**	**5**	**10,623.52**

Formula in D2: `=PMT(B2/12,C2*12,-A2)`
A 5 year loan of 500,000 at 10% a year gives about **10,623.52** a month.

5. Random Numbers and Quick Tools

- `=RAND()` gives a decimal between 0 and 1.
- `=RANDBETWEEN(1,100)` gives a whole number between 1 and 100.
- Both change every time the sheet recalculates. Paste as values to freeze them.

6. When to Use It

- Work out each product's share of total sales
- Measure growth between two months or years
- Add tax or apply a discount to a price list
- Compare monthly loan payments before borrowing

7. Important Things to Remember

- Growth is New ÷ Old - 1, then format the cell as a percentage.
- Keep PMT units matching: monthly payments need a monthly rate and months of term.
- RAND and RANDBETWEEN recalculate whenever the sheet does.

8. Common Mistakes

Mistake: Using the yearly rate in PMT for monthly payments. **Fix:** Divide the rate by 12 and multiply the years by 12.

Mistake: Getting a negative payment from PMT. **Fix:** Put a minus sign before the loan amount: `-A2`.

Mistake: Forgetting to lock the total in a share formula and getting wrong results when copying down. **Fix:** Use `$B$2:$B$6` so the range does not move.

9. Data Analyst Tip

Check the total cost of a loan: 10,623.52 × 60 months = 637,411.20. That is 137,411.20 more than the 500,000 borrowed, which is the interest you pay.

SUM Function in Excel

1. What is SUM?

The SUM function adds numbers together and gives you the total. It solves the everyday problem of totaling a list of values, such as daily sales, monthly expenses, or units in stock. A Data Analyst uses it to turn long columns of figures into totals that can be reported and compared.

2. Syntax

`=SUM(number1, [number2], ...)`

Argument	Required?	What it accepts
`number1`	Required	A number, a cell reference, or a range (for example `B2:B6`)
`number2, ...`	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

The most common form is a single range, such as `=SUM(B2:B6)`. You can also add separate ranges, `=SUM(B2:B6, D2:D6)`, or mix a range with a number, `=SUM(B2:B6, 500)`.

3. Simple Example

Find the total sales for the week:

	A	B
1	Day	Sales
2	Monday	12000
3	Tuesday	15000
4	Wednesday	10000
5	Thursday	18000
6	Friday	14000
7	**Total Sales**	**69000**

Formula in B7: `=SUM(B2:B6)`

Excel reads the five values in B2 to B6 and adds them: 12000 + 15000 + 10000 + 18000 + 14000. **Result: 69000**

4. Another Example

Find the total units sold for each product across three months:

	A	B	C	D	E
1	Product	Jan	Feb	Mar	**Total**
2	Mouse	120	150	130	**400**
3	Keyboard	80	95	75	**250**
4	Headset	60	45	55	**160**

Formula in E2: `=SUM(B2:D2)`, then copied down to E4.

For Mouse, Excel adds the cells across the row: 120 + 150 + 130. **Result: 400**. Copying the formula down gives 250 for Keyboard and 160 for Headset.

5. When to Use It

- Calculate total sales revenue for a day, month, or year
- Calculate total expenses or costs for a period
- Calculate the total number of orders or units sold
- Calculate total stock units in inventory
- Calculate row totals across months or quarters in a monthly report

6. Important Things to Remember

- SUM works with a single cell, a range, several ranges, or a mix of these. It adds down a column or across a row.
- Empty cells and text inside a range are ignored. They do not cause an error.
- Numbers stored as text (often left-aligned, sometimes with a green corner triangle) are ignored in a range, so the total comes out too low.
- If any referenced cell contains an error such as `#N/A` or `#DIV/0!`, the SUM result shows that error.
- SUM includes hidden rows and filtered-out rows in the total.
- SUM does not update itself if you add data outside the selected range. Check that the range covers all your data.

7. Common Mistakes

Mistake: Selecting an incomplete range, such as `B2:B5` when the data goes to B6. **Fix:** Check the highlighted range after typing the formula and make sure it covers every data row.

Mistake: Including the total cell in its own range, such as `=SUM(B2:B7)` typed in B7. This causes a circular reference warning. **Fix:** End the range on the last data row, above the total.

Mistake: Getting a total that is too low because some numbers are stored as text. **Fix:** Convert them to real numbers (for example with Text to Columns) and check that they right-align.

Mistake: Adding cells one by one, such as `=B2+B3+B4+B5+B6`. It is slow and easy to get wrong. **Fix:** Use a range: `=SUM(B2:B6)`.

8. Data Analyst Tip

Use the AutoSum shortcut: select the cell below your numbers and press Alt + = (Windows) to insert a SUM formula instantly. To double-check a total, select the data range and look at the Sum shown in the status bar at the bottom of the Excel window. It should match your formula result.

AVERAGE Function in Excel

1. What is AVERAGE?

The AVERAGE function adds up a set of numbers and divides the total by how many numbers there are. The result is the arithmetic mean. It solves the problem of finding a "typical" or central value in a list, such as the average score, sales amount, or salary. A Data Analyst uses it to summarize large columns of data into one meaningful number.

2. Syntax

`=AVERAGE(number1, [number2], ...)`

Argument	Required?	What it accepts
`number1`	Required	A number, a cell reference, or a range (for example `B2:B5`)
`number2, ...`	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

Most often you will use a single range, such as `=AVERAGE(B2:B5)`. You can also combine separate ranges: `=AVERAGE(B2:B5, D2:D5)`.

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	70
3	Karim	80
4	Hasan	60
5	Nabil	90
6	**Average**	**75**

Formula: `=AVERAGE(B2:B5)`

Excel reads 70, 80, 60, 90, adds them (70 + 80 + 60 + 90 = 300), counts them (4), and divides: 300 ÷ 4. **Result: 75**

4. Another Example

	A	B
1	Product	Monthly Sales
2	Laptop	12000
3	Mouse	3500
4	Keyboard	4500
5	Monitor	8000
6	Headset	(blank)
7	**Average**	**7000**

Formula: `=AVERAGE(B2:B6)`

Excel adds 12000 + 3500 + 4500 + 8000 = 28000 and divides by 4, because the blank cell B6 is ignored. **Result: 7000**

5. When to Use It

- Calculate the average exam score of a class
- Calculate average monthly sales per product
- Calculate average employee salary in a department
- Calculate average customer spending per order
- Calculate average daily expenses or marketing cost per campaign

6. Important Things to Remember

- AVERAGE = SUM ÷ COUNT. The count includes only the numbers Excel actually used.
- Text and empty cells inside a range are ignored. They are not counted as zero.
- A cell containing the number 0 is not ignored. It is counted and pulls the average down.
- If a referenced cell contains an error such as `#N/A` or `#DIV/0!`, the whole AVERAGE result shows that error.
- Numbers stored as text (often left-aligned, sometimes with a green corner triangle) are ignored in a range. Convert them to real numbers first.
- If the range has no numbers at all, AVERAGE returns `#DIV/0!`.

7. Common Mistakes

Mistake: Selecting the wrong or incomplete range, such as leaving out the last row. **Fix:** Check the highlighted range after typing the formula, or select the data with Ctrl + Shift + Down Arrow.

Mistake: Including the header or a Total row in the range. **Fix:** Select only the data rows. A Total row would be counted as a data point and distort the average.

Mistake: Confusing a blank cell with a zero. A blank is ignored, but a 0 is counted. **Fix:** Decide whether missing values should count as zero, and enter 0 only when that is what you mean.

Mistake: Getting the wrong result because numbers are stored as text. **Fix:** Convert them to numbers (for example with Text to Columns or by re-entering them) before averaging.

8. Data Analyst Tip

Always compare the AVERAGE with the data it came from. Before trusting an average, check how many numbers were actually used by calculating `=COUNT(B2:B6)` next to it. If the count is lower than the number of rows, some values are blank or stored as text, and your average may not represent the full dataset.

COUNT Function in Excel

1. What is COUNT?

The COUNT function counts how many cells contain numbers. It solves the problem of finding out how many records have a numeric value, such as how many employees have a sales figure entered. A Data Analyst uses it to check data completeness and to know how many valid numbers sit behind a total or an average. Dates and times are stored as numbers, so COUNT counts them too.

2. Syntax

`=COUNT(value1, [value2], ...)`

Argument	Required?	What it accepts
`value1`	Required	A number, a date, a cell reference, or a range (for example `B2:B5`)
`value2, ...`	Optional	More numbers, dates, cells, or ranges. Excel accepts up to 255 arguments.

Most often you will use a single range, such as `=COUNT(B2:B5)`. The result is the number of cells that contain numbers, not the total of those numbers.

3. Simple Example

Count the employees who have a numeric sales figure:

	A	B
1	Employee	Sales
2	Rahim	25000
3	Karim	42000
4	Hasan	(blank)
5	Nabil	50000
6	**Count**	**3**

Formula in B6: `=COUNT(B2:B5)`

Excel checks each cell in B2:B5. Three cells contain numbers (25000, 42000, 50000). The blank cell B4 is skipped. **Result: 3**

4. Another Example

Count how many students have a mark entered:

	A	B
1	Student	Marks
2	Aisha	85
3	Rahim	Absent
4	Nabila	0
5	Tanvir	(blank)
6	Sabbir	72
7	Mitu	90
8	**Count**	**4**

Formula in B8: `=COUNT(B2:B7)`

Excel counts 85, 0, 72, and 90. The text "Absent" and the blank cell are ignored, but the 0 is counted because it is a number. **Result: 4**

5. When to Use It

- Count how many employees have a sales figure recorded
- Count how many exam marks have been entered
- Count how many transactions have a recorded amount
- Check how many numeric values are in a column before averaging or totaling it
- Count how many dates are filled in, such as order dates

6. Important Things to Remember

- COUNT counts only cells that contain numbers. Dates and times count because Excel stores them as numbers.
- Text and blank cells inside a range are ignored.
- A cell containing 0 is counted, because 0 is a number. A blank cell is not.
- Numbers stored as text (often left-aligned, sometimes with a green corner triangle) are not counted in a range.
- Error values such as `#N/A` inside the range are ignored. COUNT does not return an error.
- COUNT tells you how many numbers there are, not their total. To count all non-empty cells, including text, use COUNTA.

7. Common Mistakes

Mistake: Expecting COUNT to count text entries such as names or statuses. **Fix:** COUNT only counts numbers. Use COUNTA to count non-empty cells of any type.

Mistake: Getting a count that is too low because numbers are stored as text. **Fix:** Convert them to real numbers (for example with Text to Columns) and check that they right-align.

Mistake: Selecting an incomplete range, such as `B2:B4` when data goes to B5. **Fix:** Check the highlighted range after typing the formula.

Mistake: Using COUNT when the total is needed. **Fix:** COUNT answers "how many", SUM answers "how much".

8. Data Analyst Tip

Use COUNT as a quick data quality check. If a column should have 500 values and `=COUNT(B2:B501)` returns 480, then 20 cells are blank or contain text. Fix those before you build totals and averages. This also lets you report results like "480 of 500 employees have sales recorded."

IF Function in Excel

1. What is IF?

The IF function tests a condition and returns one result if the condition is TRUE and a different result if it is FALSE. It solves the problem of making a decision for every row of data, such as labelling a student as Pass or Fail. A Data Analyst uses it to categorize records, flag important rows, and create new columns from existing data.

2. Syntax

`=IF(logical_test, value_if_true, value_if_false)`

Argument	Required?	What it accepts
`logical_test`	Required	A condition that is either TRUE or FALSE, such as `B2>=50` or `C2="Sales"`
`value_if_true`	Optional*	What to return when the test is TRUE: text in quotes (`"Pass"`), a number, a cell, or a calculation
`value_if_false`	Optional*	What to return when the test is FALSE: the same types as above

*Excel allows these two to be left out, but beginners should always fill in both. Comparison operators used in the test: `=` equal, `<>` not equal, `>` greater, `<` less, `>=` greater or equal, `<=` less or equal.

3. Simple Example

Check whether each student passed (pass mark is 50):

	A	B	C
1	Student	Marks	**Result**
2	Rahim	75	**Pass**
3	Karim	45	**Fail**
4	Hasan	82	**Pass**
5	Nabil	55	**Pass**

Formula in C2: `=IF(B2>=50, "Pass", "Fail")`, then copied down to C5.

For Rahim, Excel checks whether 75 is 50 or more. This is TRUE, so it returns "Pass". For Karim, 45 is not 50 or more. This is FALSE, so it returns "Fail". Excel repeats this for each row.

4. Another Example

Pay a 5% bonus on sales of 10000 or more, otherwise 0:

	A	B	C
1	Employee	Sales	**Bonus**
2	Anika	12000	**600**
3	Sabbir	8500	**0**
4	Mitu	15000	**750**
5	Tanvir	9800	**0**

Formula in C2: `=IF(B2>=10000, B2*5%, 0)`, then copied down to C5.

For Anika, 12000 is 10000 or more, so Excel returns the calculation 12000 × 5% = 600. For Sabbir, 8500 is below 10000, so Excel returns 0.

5. When to Use It

- Label exam marks as Pass or Fail
- Check whether a sales target was met or not met
- Decide bonus eligibility based on sales or performance
- Mark attendance as Eligible or Not Eligible
- Flag high-value orders or overdue payments for review

6. Important Things to Remember

- IF checks one condition and returns one of two results: the TRUE result or the FALSE result.
- Text results must be inside quotation marks (`"Pass"`). Numbers, cell references, and calculations do not need quotes.
- The results can be calculations, not just labels, for example `B2*5%`.
- If you leave out `value_if_false`, as in `=IF(B2>=50,"Pass")`, Excel shows FALSE for failing rows. Always provide both results.
- Text comparisons are not case-sensitive: `C2="sales"` matches "Sales".
- IF works on one row at a time. Write it in the first row, then copy it down the column. For more than two outcomes, one IF can be placed inside another.

7. Common Mistakes

Mistake: Forgetting quotation marks around a text result, which can cause a `#NAME?` error. **Fix:** Write `"Pass"` and `"Fail"` with quotes.

Mistake: Using the wrong operator at the boundary, such as `>50` instead of `>=50`, so a student with exactly 50 fails. **Fix:** Decide whether the boundary value should count, then choose `>` or `>=`.

Mistake: Putting a number in quotes inside the test, such as `B2>="50"`. Excel treats it as text and the result is wrong. **Fix:** Write numbers without quotes: `B2>=50`.

Mistake: Missing or extra brackets and commas. **Fix:** Check the formula bar; Excel colors each pair of brackets to help you see which ones match.

8. Data Analyst Tip

Put the condition value in its own cell instead of typing it inside the formula. For example, store the pass mark 50 in E1 and use `=IF(B2>=$E$1, "Pass", "Fail")`. If the pass mark changes, update E1 once and every result updates. IF is also useful for creating a status column that you can then filter, sort, or count.

IFS Function in Excel

1. What is IFS?

The IFS function checks several conditions in order and returns the result for the first condition that is TRUE. It solves problems where a value must be placed into one of many categories, such as grades, rating levels, or commission bands. A Data Analyst uses it to create labels and categories without writing IF formulas inside each other.

2. Syntax

`=IFS(logical_test1, value_if_true1, [logical_test2, value_if_true2], ...)`

Argument	Required?	What it accepts
`logical_test1`	Required	The first condition, such as `B2>=90`
`value_if_true1`	Required	The result if the first condition is TRUE: text in quotes, a number, or a calculation
`logical_test2, value_if_true2`	Optional	More condition and result pairs. Excel accepts up to 127 pairs.

Every condition must be followed by its own result. Excel checks the pairs from left to right and stops at the first TRUE condition. IFS is available in Excel 2019, Excel 2021, and Microsoft 365, but not in Excel 2016 or earlier.

3. Simple Example

Assign a performance level based on score:

	A	B	C
1	Student	Score	**Performance**
2	Aisha	92	**Excellent**
3	Rahim	78	**Good**
4	Nabila	65	**Average**
5	Tanvir	48	**Needs Improvement**

Formula in C2, copied down to C5:

```
=IFS(B2>=90,"Excellent", B2>=75,"Good", B2>=60,"Average", B2<60,"Needs Improvement")
```

For Rahim (78): 78>=90 is FALSE, then 78>=75 is TRUE, so Excel returns "Good" and stops. For Tanvir (48): the first three tests are FALSE and 48<60 is TRUE, so he gets "Needs Improvement".

4. Another Example

Label stock levels: 0 is Out of Stock, 20 or less is Low, anything else is In Stock:

	A	B	C
1	Product	Stock	**Status**
2	Mouse	0	**Out of Stock**
3	Keyboard	15	**Low**
4	Monitor	60	**In Stock**
5	Headset	20	**Low**

Formula in C2, copied down to C5:

```
=IFS(B2=0,"Out of Stock", B2<=20,"Low", TRUE,"In Stock")
```

For Mouse (0): the first test is TRUE, so Excel returns "Out of Stock". For Headset (20): the first test is FALSE and 20<=20 is TRUE, so it returns "Low". For Monitor (60): both tests are FALSE, so the final TRUE test returns "In Stock".

5. When to Use It

- Convert exam scores into grades (A, B, C, D, F)
- Rate employee performance as Excellent, Good, Average, or Needs Improvement
- Apply different sales commission rates based on the sales amount
- Label inventory as Out of Stock, Low, or In Stock
- Group customers into tiers based on total spending

6. Important Things to Remember

- IFS checks conditions from left to right and returns the result of the first TRUE condition. It ignores the rest.
- Order matters. With `>=` tests, go from the highest value to the lowest, or lower tests will catch high values first.
- Each condition needs its own result. A missing result causes an error.
- If no condition is TRUE, IFS returns `#N/A`. Use `TRUE` as the last condition to handle all remaining cases.
- Text results must be in quotation marks. Numbers and calculations do not need quotes.
- Use IF for one condition (two outcomes) and IFS for three or more outcomes.

7. Common Mistakes

Mistake: Putting the conditions in the wrong order, such as `B2>=60` before `B2>=90`, so a score of 92 returns "Average". **Fix:** List `>=` conditions from highest to lowest.

Mistake: Getting `#N/A` because no condition was TRUE. **Fix:** Add `TRUE, "result"` as the last pair, or make sure your conditions cover every possible value.

Mistake: Leaving out a result or a comma, so the conditions and results do not form pairs. **Fix:** Read the formula as pairs: condition, result, condition, result.

Mistake: Forgetting quotation marks around text results. **Fix:** Write results like `"Good"` inside quotes.

8. Data Analyst Tip

After filling an IFS formula down a column, test the boundary values (for example 90, 75, and 60) to confirm each lands in the correct category. Then filter the new column to check for any `#N/A` results. This quickly shows gaps in your conditions.

SUMIF Function in Excel

1. What is SUMIF?

The SUMIF function adds only those numbers that meet one condition. It solves the problem of finding a total for one group inside a larger list, such as total sales for one region or total cost for one expense category. A Data Analyst uses it to build group totals without filtering or copying the data.

2. Syntax

`=SUMIF(range, criteria, [sum_range])`

Argument	Required?	What it accepts
`range`	Required	The cells Excel checks against the condition (for example `B2:B10`)
`criteria`	Required	The condition to match: text (`"North"`), a number (`100`), a comparison (`">=500"`), or a cell reference
`sum_range`	Optional	The cells that are actually added (for example `C2:C10`). If left out, Excel adds the cells in `range` itself.

Use the third argument when the condition is in one column and the numbers to add are in another. If both are the same column, you can leave it out: `=SUMIF(B2:B7, ">=500")`.

3. Simple Example

Find the total sales for the North region:

	A	B	C
1	Product Category	Region	Sales
2	Laptop	North	5000
3	Laptop	South	7000
4	Laptop	East	6000
5	Monitor	North	3000
6	Monitor	South	4500
7	Monitor	East	3800
8	Mouse	North	2500
9	Mouse	South	3200
10	Mouse	East	2800
11	**Total**	**North**	**10500**

Formula in C11: `=SUMIF(B2:B10, "North", C2:C10)`

Excel looks for "North" in B2:B10 (rows 2, 5, 8), takes the matching sales from C2:C10 (5000, 3000, 2500), and adds them. **Result: 10500**

4. Another Example

Find the total of expenses that are 500 or more:

	A	B
1	Expense	Amount
2	Rent	1200
3	Internet	300
4	Travel	650
5	Supplies	150
6	Software	500
7	Snacks	90
8	**Total (>=500)**	**2350**

Formula in B8: `=SUMIF(B2:B7, ">=500")`

Here the condition and the numbers to add are in the same column, so no third argument is needed. Excel finds the amounts of 500 or more (1200, 650, 500) and adds them. **Result: 2350**

5. When to Use It

- Calculate total sales for one region or product
- Calculate total expenses for one category
- Calculate the total value of orders with a given status
- Calculate total units in stock for one product
- Calculate total hours or performance points for one team

6. Important Things to Remember

- SUMIF works with **one** condition only. For two or more conditions, use SUMIFS.
- Text and comparison criteria go inside quotation marks: `"North"`, `">=500"`, `"<>East"`. Matching is not case-sensitive.
- `range` and `sum_range` must be the same size, because Excel matches them row by row.
- Text and blank cells in `sum_range` are ignored.
- If no cell meets the condition, SUMIF returns 0. This can hide a spelling problem, so check unexpected zeros.
- Wildcards work with text: `"Lap*"` matches any text starting with Lap. To use a cell value in a comparison, join it: `">="&E1`.

7. Common Mistakes

Mistake: Forgetting quotation marks around text or comparison criteria, such as `North` or `>=500`. **Fix:** Write `"North"` and `">=500"`.

Mistake: Using ranges of different sizes, such as `B2:B10` with `C2:C9`. **Fix:** Select the same rows in both ranges.

Mistake: Swapping the order and putting the numbers first. **Fix:** Remember the order: condition range, condition, then numbers to add.

Mistake: Getting 0 because the text does not match exactly, such as "North " with an extra space. **Fix:** Clean the data first, or check the unique values in the column.

8. Data Analyst Tip

Type each group name in a cell and refer to it, for example `=SUMIF($B$2:$B$10, E2, $C$2:$C$10)`, then copy the formula down to build a summary by region. Always check your work: the group totals should add up to `=SUM(C2:C10)`. Here 10500 + 14700 + 12600 = 37800. If they do not match, a region name is misspelled or missing.

COUNTIF Function in Excel

1. What is COUNTIF?

The COUNTIF function counts how many cells meet one condition. It solves the problem of finding out how many items belong to a group, such as how many employees work in Marketing or how many orders are above 100. A Data Analyst uses it to count records by category, status, or value without filtering the data.

2. Syntax

`=COUNTIF(range, criteria)`

Argument	Required?	What it accepts
`range`	Required	The cells Excel checks against the condition (for example `B2:B9`)
`criteria`	Required	The condition to match: text (`"Marketing"`), a number (`100`), a comparison (`">100"`), or a cell reference

COUNTIF has no third argument. It counts the cells in `range` that match the criteria; it does not add or average anything. Comparison operators: `>` greater, `<` less, `>=` greater or equal, `<=` less or equal, `=` equal, `<>` not equal.

3. Simple Example

Count how many employees are in the Marketing department:

	A	B
1	Employee	Department
2	Aisha	Marketing
3	Rahim	Finance
4	Nabila	Analytics
5	Tanvir	Operations
6	Farhan	Marketing
7	Sadia	Finance
8	Rafiq	Marketing
9	Jannat	Analytics
10	**Count**	**3**

Formula in B10: `=COUNTIF(B2:B9, "Marketing")`

Excel checks each cell in B2:B9 and counts the ones equal to "Marketing" (Aisha, Farhan, Rafiq). **Result: 3**

4. Another Example

Count the products with low stock (less than 20 units):

	A	B
1	Product	Stock
2	Mouse	45
3	Keyboard	12
4	Monitor	8
5	Headset	30
6	Webcam	19
7	Cable	60
8	Laptop	5
9	**Count (<20)**	**4**

Formula in B9: `=COUNTIF(B2:B8, "<20")`

Excel checks each stock value and counts those below 20: 12, 8, 19, and 5. The result is how many products meet the condition, not their total. **Result: 4**

5. When to Use It

- Count employees in each department
- Count high-value sales or orders above a certain amount
- Count low-stock items in inventory
- Count customers by city, status, or type
- Count campaign responses, such as how many answered "Yes"

6. Important Things to Remember

- COUNTIF works with **one** condition only. For two or more conditions, use COUNTIFS.
- Text and comparison criteria go inside quotation marks: `"Marketing"`, `">100"`, `"<>Finance"`. Matching is not case-sensitive.
- The result is a count of matching cells, not a total of their values.
- If no cell meets the condition, COUNTIF returns 0, not an error.
- Wildcards work with text: `"Mark*"` matches any text starting with Mark. To use a cell value in a comparison, join it: `">"&E1`.
- The criteria can be a cell reference, so one formula can be copied for many groups.

7. Common Mistakes

Mistake: Forgetting quotation marks around text or comparison criteria, such as `Marketing` or `>100`. **Fix:** Write `"Marketing"` and `">100"`.

Mistake: Getting 0 because the text does not match exactly, such as "Marketing " with an extra space or a spelling difference. **Fix:** Clean the data first, or check the unique values in the column.

Mistake: Copying the formula down without locking the range, so it shifts to the wrong rows. **Fix:** Use absolute references, such as `$B$2:$B$9`.

Mistake: Trying to count with two conditions in one COUNTIF. **Fix:** Use COUNTIFS when you need to meet more than one condition.

8. Data Analyst Tip

Type each group name in a cell and use it as the criteria, for example `=COUNTIF($B$2:$B$9, D2)`, then copy the formula down to build a count by department. Always check your work: the group counts should add up to the number of records. Here 3 + 2 + 2 + 1 = 8 employees. If they do not, a department name is misspelled or missing.

AVERAGEIF Function in Excel

1. What is AVERAGEIF?

The AVERAGEIF function calculates the average of only those cells that meet one condition. It solves the problem of finding the average for one group inside a larger list, such as the average salary of the Sales department or the average score of students who passed. A Data Analyst uses it to get group-level averages without filtering or copying the data.

2. Syntax

`=AVERAGEIF(range, criteria, [average_range])`

Argument	Required?	What it accepts
`range`	Required	The cells Excel checks against the condition (for example `B2:B9`)
`criteria`	Required	The condition to match: text (`"Sales"`), a number (`100`), a comparison (`">=85"`), or a cell reference
`average_range`	Optional	The cells that are actually averaged (for example `C2:C9`). If left out, Excel averages `range` itself.

Use the third argument when the condition is in one column and the numbers to average are in another. If both are the same column, you can leave it out: `=AVERAGEIF(C2:C8, ">=85")`.

3. Simple Example

	A	B	C
1	Employee ID	Department	Salary
2	E001	IT	4800
3	E002	Sales	5200
4	E003	Marketing	4600
5	E004	Sales	6100
6	E005	HR	4300
7	E006	Sales	4700
8	E007	IT	5500
9	E008	Sales	5300
10	**Average**	**Sales**	**5325**

Formula: `=AVERAGEIF(B2:B9, "Sales", C2:C9)`

Excel checks B2:B9 for "Sales" (rows 3, 5, 7, 9), takes the matching salaries from C2:C9 (5200, 6100, 4700, 5300), adds them (21300), and divides by 4. **Result: 5325**

4. Another Example

Average delivery days for orders worth more than 100:

	A	B	C
1	Order	Order Value	Delivery Days
2	O1	250	3
3	O2	80	5
4	O3	120	2
5	O4	60	4
6	O5	300	4
7	O6	90	6
8	**Average**	**>100**	**3**

Formula: `=AVERAGEIF(B2:B7, ">100", C2:C7)`

Excel finds the orders above 100 (O1, O3, O5), takes their delivery days (3, 2, 4), adds them (9), and divides by 3. Orders worth 80, 60 and 90 are left out. **Result: 3**

5. When to Use It

- Calculate the average salary of one department
- Calculate average sales for a single region or product
- Calculate the average score of students who reached a pass mark
- Calculate average order value for orders above a certain amount
- Calculate average delivery time or expense for one category

6. Important Things to Remember

- AVERAGEIF works with **one** condition only. For two or more conditions, use AVERAGEIFS.
- Text and comparison criteria go inside quotation marks: `"Sales"`, `">100"`, `"<>HR"`. Matching is not case-sensitive.
- `range` and `average_range` must be the same size, because Excel matches them row by row.
- Blank cells and text in `average_range` are ignored. If `average_range` is left out, `range` must contain the numbers.
- If no cell meets the condition, AVERAGEIF returns `#DIV/0!`.
- Wildcards work with text: `"S*"` matches any text starting with S. To use a cell value in a comparison, join it: `">="&E1`.

7. Common Mistakes

Mistake: Forgetting quotation marks around text or comparison criteria, such as `Sales` or `>100`. **Fix:** Write `"Sales"` and `">100"`.

Mistake: Using ranges of different sizes, such as `B2:B9` with `C2:C10`. **Fix:** Select the same rows in both ranges.

Mistake: Swapping the order, putting the numbers first and the condition column last. **Fix:** Remember the order: condition range, condition, then numbers to average.

Mistake: Text that does not match exactly, such as "Sales " with an extra space or a spelling difference. **Fix:** Clean the data first, or check the unique values in the column.

8. Data Analyst Tip

Type the criteria in a cell and refer to it instead of typing it inside the formula. For example, put "Sales" in E2 and use `=AVERAGEIF($B$2:$B$9, E2, $C$2:$C$9)`. Then list every department in column E and copy the formula down to build a full average-by-department summary in seconds.

TEXTJOIN Function in Excel

1. What is TEXTJOIN?

The TEXTJOIN function joins text from several cells or a whole range, puts a separator of your choice between the items, and can skip empty cells. It solves the problem of combining many values into one clean piece of text, such as a list of names or a full address. A Data Analyst uses it to merge columns and build lists without extra commas or gaps from blank cells.

2. Syntax

`=TEXTJOIN(delimiter, ignore_empty, text1, [text2], ...)`

Argument	Required?	What it accepts
`delimiter`	Required	The separator placed between items, in quotes: `", "`, `" - "`, or `""` for no separator. A cell reference also works.
`ignore_empty`	Required	`TRUE` skips empty cells. `FALSE` keeps them. Do not put quotes around TRUE or FALSE.
`text1`	Required	The text, cell, or range to join (for example `A2:C2`)
`text2, ...`	Optional	More text, cells, or ranges to join

The delimiter appears only between items, never at the start or end. Unlike CONCAT, TEXTJOIN adds the separator for you, so you do not repeat it in the formula.

3. Simple Example

Join names with a comma, ignoring blank cells:

	A	B	C	D
1	Name 1	Name 2	Name 3	**Result**
2	Rahim	Khan	Ali	**Rahim, Khan, Ali**
3	Karim		Hasan	**Karim, Hasan**
4	Nabil	Hossain		**Nabil, Hossain**
5	Sadia	Rahman	Rafi	**Sadia, Rahman, Rafi**

Formula in D2: `=TEXTJOIN(", ", TRUE, A2:C2)`, then copied down to D5.

Excel joins the three cells in A2:C2 with ", " between them. In row 3, the blank cell B3 is skipped because ignore_empty is TRUE. **Result: Karim, Hasan**

4. Another Example

Put all products from a list into one cell (row 4 is blank):

	A
1	Product
2	Mouse
3	Keyboard
4	
5	Monitor
6	Headset
7	**Mouse, Keyboard, Monitor, Headset**

Formula in A7: `=TEXTJOIN(", ", TRUE, A2:A6)`

Excel joins the products down the column in order. The blank cell A4 is skipped. With FALSE instead of TRUE, the result would have an extra comma: "Mouse, Keyboard, , Monitor, Headset".

5. When to Use It

- Merge name parts into one name, skipping any that are blank
- Build a comma-separated list of products, customers, or items in one cell
- Build an address from street, area, and city when some parts are missing
- Create labels such as "Product - Category - Price" from several columns
- Combine tags, skills, or notes from multiple columns into one

6. Important Things to Remember

- The first argument is the delimiter and the second is `TRUE` or `FALSE`. Both must come before the text.
- `TRUE` skips empty cells. `FALSE` keeps them, which creates extra delimiters, such as "Karim, , Hasan".
- TEXTJOIN accepts ranges across a row or down a column, so you do not have to list each cell.
- The result is text. Numbers are joined as stored: 1200 stays 1200, not 1,200.
- The result cannot be longer than 32,767 characters, or Excel shows `#VALUE!`.
- TEXTJOIN is available in Excel 2019, Excel 2021, and Microsoft 365, but not in Excel 2016 or earlier.

7. Common Mistakes

Mistake: Leaving out the `ignore_empty` argument, such as `=TEXTJOIN(", ", A2:C2)`. **Fix:** Always include `TRUE` or `FALSE` as the second argument.

Mistake: Using `FALSE` by accident, so blank cells leave extra commas in the result. **Fix:** Use `TRUE` unless you want blank cells to be kept.

Mistake: Typing the delimiter without quotation marks, such as `=TEXTJOIN(-, TRUE, A2:C2)`. **Fix:** Write the delimiter in quotes: `"-"`.

Mistake: Joining dates and seeing serial numbers such as 46023. **Fix:** Convert the date with the TEXT function first, for example `TEXT(A2, "dd-mmm-yyyy")`.

8. Data Analyst Tip

Use TEXTJOIN with a whole column to build a quick list for a report or an email, for example `=TEXTJOIN(", ", TRUE, A2:A50)`. Keep ignore_empty set to TRUE so that blank rows do not create extra commas. If you need to keep the result after changing the source data, copy it and paste as values.

CONCAT Function in Excel

1. What is CONCAT?

The CONCAT function joins text, numbers, or cell values into one combined piece of text. It solves the problem of having related information in separate columns, such as a first name and a last name. A Data Analyst uses it to build full names, IDs, addresses, and labels from existing columns without retyping.

2. Syntax

`=CONCAT(text1, [text2], ...)`

Argument	Required?	What it accepts
`text1`	Required	Text in quotes, a number, a cell reference, or a range (for example `A2` or `A2:C2`)
`text2, ...`	Optional	More text, numbers, cells, or ranges to join. Excel accepts up to 253 arguments.

CONCAT puts nothing between the items. To add a space or a symbol, type it as text in quotes, such as `" "` or `"-"`. Using a range, such as `=CONCAT(A2:C2)`, joins every cell in that range one after another with no separator.

3. Simple Example

Combine first name and last name with a space:

	A	B	C
1	First Name	Last Name	**Full Name**
2	Rahim	Khan	**Rahim Khan**
3	Karim	Ali	**Karim Ali**
4	Hasan	Rana	**Hasan Rana**
5	Nabil	Hossain	**Nabil Hossain**

Formula in C2: `=CONCAT(A2, " ", B2)`, then copied down to C5.

For row 2, Excel takes the text in A2 ("Rahim"), adds the space from the quotes, then adds the text in B2 ("Khan"). **Result: Rahim Khan**

4. Another Example

Create an order ID from a region code and an order number:

	A	B	C
1	Region Code	Order No	**Order ID**
2	DHK	1001	**DHK-1001**
3	CTG	1002	**CTG-1002**
4	SYL	1003	**SYL-1003**
5	RAJ	1004	**RAJ-1004**

Formula in C2: `=CONCAT(A2, "-", B2)`, then copied down to C5.

Excel joins the code in A2, a hyphen, and the number in B2. The result is text, even though part of it came from a number. **Result: DHK-1001**

5. When to Use It

- Create full names from separate first name and last name columns
- Create IDs or codes by joining fields, such as region code and number
- Build addresses from street, city, and ZIP columns
- Create labels such as "Product - Category" for reports
- Create a unique key from two columns to use in lookups

6. Important Things to Remember

- CONCAT adds nothing between items by itself. Add spaces and symbols as text in quotes.
- Typed text must be in quotation marks. Cell references do not need quotes.
- The result is always text, even if you join numbers. It cannot be used directly in calculations as a number.
- Numbers are joined as stored: 1200 stays 1200, not 1,200. Dates appear as serial numbers, such as 46023.
- CONCAT accepts ranges such as `A2:C2`. The older CONCATENATE function does not.
- CONCAT is available in Excel 2019, Excel 2021, and Microsoft 365, but not in Excel 2016 or earlier.

7. Common Mistakes

Mistake: Forgetting the space, so "Rahim" and "Khan" become "RahimKhan". **Fix:** Add `" "` between the two cells.

Mistake: Forgetting quotation marks around typed text, such as `=CONCAT(A2, -, B2)`. **Fix:** Write `"-"` in quotes.

Mistake: Joining a date and getting a number such as 46023. **Fix:** Convert the date to text first with the TEXT function, for example `TEXT(A2, "dd-mmm-yyyy")`.

Mistake: Using a range such as `=CONCAT(A2:C2)` and expecting spaces between the items. **Fix:** Range joining adds no separators. List each cell and separator individually.

8. Data Analyst Tip

Use CONCAT to build a unique key when two columns together identify a record, for example `=CONCAT(A2, "-", B2)`. This makes lookups and duplicate checks easier. If you need to keep the joined results after deleting the original columns, copy them and paste as values first.

LEFT, RIGHT and MID in Excel

1. What are LEFT, RIGHT and MID?

LEFT, RIGHT, and MID extract a set number of characters from text. LEFT takes them from the start, RIGHT takes them from the end, and MID takes them from any position in the middle. They solve the problem of codes and IDs that combine several pieces of information in one cell, such as a prefix, a year, and a number. A Data Analyst uses them to split these codes into separate columns.

2. Syntax

Function	Syntax	Arguments
LEFT	`=LEFT(text, [num_chars])`	`text` is required. `num_chars` is optional. The default is 1.
RIGHT	`=RIGHT(text, [num_chars])`	`text` is required. `num_chars` is optional. The default is 1.
MID	`=MID(text, start_num, num_chars)`	`text`, `start_num`, and `num_chars` are all required.

`text` is a cell reference or text in quotes. `num_chars` is how many characters to return. `start_num` is the position of the first character to take, where the first character in the text is position 1. Spaces count as characters.

3. Simple Example

Cell A2 contains the word Bangladesh. Each letter has a position:

Letter	B	a	n	g	l	a	d	e	s	h
Position	1	2	3	4	5	6	7	8	9	10

Formula	Result
`=LEFT(A2, 3)`	**Ban**
`=RIGHT(A2, 3)`	**esh**
`=MID(A2, 4, 3)`	**gla**

LEFT takes the first 3 characters. RIGHT takes the last 3. MID starts at position 4 (g) and takes 3 characters.

4. Another Example

Split employee IDs into a prefix, a year, and a number:

	A	B	C	D
1	Employee ID	**Prefix**	**Year**	**No.**
2	EMP-2026-105	**EMP**	**2026**	**105**
3	EMP-2025-212	**EMP**	**2025**	**212**
4	EMP-2026-087	**EMP**	**2026**	**087**

Formulas in row 2, copied down: B2 `=LEFT(A2, 3)`, C2 `=MID(A2, 5, 4)`, D2 `=RIGHT(A2, 3)`

In EMP-2026-105, the year starts at position 5 and is 4 characters long. The results are text, so the leading zero in "087" is kept.

5. When to Use It

- Extract prefixes or country codes from the start of IDs (LEFT)
- Extract the last digits of an ID, or a postal code from the end (RIGHT)
- Extract a year, department code, or other part from the middle of a code (MID)
- Split fixed-format codes into separate columns
- Extract a file extension or an area code that always has the same length

6. Important Things to Remember

- LEFT starts at the first character, RIGHT starts at the last character, and MID starts at any position you choose.
- `num_chars` is optional in LEFT and RIGHT (default 1), but MID needs all three arguments. MID has no default.
- The results are text, even when they look like numbers, such as "105" or "087". Use `VALUE()` to convert them to numbers.
- If `num_chars` is longer than the text, you get the whole text. In MID, a `start_num` beyond the end gives an empty result.
- A `start_num` below 1 or a negative `num_chars` gives `#VALUE!`.
- These functions work best when every code has the same length and layout.

7. Common Mistakes

Mistake: Leaving out arguments in MID, such as `=MID(A2)`, which gives an error. **Fix:** Always give `start_num` and `num_chars`: `=MID(A2, 5, 4)`.

Mistake: Counting positions wrongly, for example forgetting that the first character is position 1, or skipping spaces and hyphens. **Fix:** Write out the characters with their positions, as in the Bangladesh example.

Mistake: Adding up the extracted numbers and getting 0, because they are text. **Fix:** Convert them with `VALUE(RIGHT(A2, 3))`.

Mistake: Using a fixed number of characters on text of different lengths, so words are cut off. **Fix:** Find the position first with FIND, as in the tip below.

8. Data Analyst Tip

When the pieces have different lengths, let Excel find the position for you. For example, `=LEFT(A2, FIND("-", A2) - 1)` returns everything before the first hyphen, whatever its length. For EMP-2026-105 it returns EMP.

MAX Function in Excel

1. What is MAX?

The MAX function returns the largest number in a set of values. It solves the problem of finding the highest value in a long list, such as the highest sale, the top score, or the biggest salary. A Data Analyst uses it to find peak values and to compare other figures against the best result.

2. Syntax

`=MAX(number1, [number2], ...)`

Argument	Required?	What it accepts
`number1`	Required	A number, a cell reference, or a range (for example `B2:B5`)
`number2, ...`	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

Most often you will use a single range, such as `=MAX(B2:B5)`. You can also mix a range with a number: `=MAX(B2:B5, 60000)` returns whichever is larger, the highest value in the range or 60000.

3. Simple Example

Find the highest sales figure:

	A	B
1	Employee	Sales
2	Rahim	25000
3	Karim	42000
4	Hasan	35000
5	Nabil	50000
6	**Maximum**	**50000**

Formula in B6: `=MAX(B2:B5)`

Excel compares the four values in B2 to B5 (25000, 42000, 35000, 50000) and returns the largest one. **Result: 50000**

4. Another Example

Find the highest mark. One student was absent and one mark is missing:

	A	B
1	Student	Marks
2	Aisha	82
3	Rahim	Absent
4	Nabila	91
5	Tanvir	(blank)
6	Sadia	77
7	**Highest**	**91**

Formula in B7: `=MAX(B2:B6)`

Excel ignores the text "Absent" and the blank cell, and compares only 82, 91, and 77. **Result: 91**

5. When to Use It

- Find the highest sales figure or the best-selling product value
- Find the top exam score in a class
- Find the highest salary in a department
- Find the highest temperature, price, or expense in a period
- Find the most recent date in a list (dates are stored as numbers)

6. Important Things to Remember

- Text and empty cells inside a range are ignored. They are not counted as zero.
- A cell containing 0 is counted. Among negative numbers, the largest is the one closest to zero, so -5 is larger than -20.
- If a referenced cell contains an error such as `#N/A`, the whole MAX result shows that error.
- Numbers stored as text (often left-aligned) are ignored in a range. Convert them to real numbers first.
- For dates, MAX returns the latest date. Format the result cell as a date to read it.
- If the range has no numbers at all, MAX returns 0.

7. Common Mistakes

Mistake: Selecting an incomplete range, such as leaving out the last row. **Fix:** Check the highlighted range after typing the formula.

Mistake: Including a Total row in the range, so MAX returns the total. **Fix:** Select only the data rows.

Mistake: Getting a wrong maximum because some numbers are stored as text. **Fix:** Convert them to numbers (for example with Text to Columns) and check that they right-align.

Mistake: Expecting MAX to show who has the highest value. **Fix:** MAX returns only the number. Use a lookup to find the name (see the tip below).

8. Data Analyst Tip

MAX gives the value, not the name. To find who has it, combine it with a lookup: `=INDEX(A2:A5, MATCH(MAX(B2:B5), B2:B5, 0))` returns Nabil for the first example. It is also good practice to check your range with `=COUNT(B2:B5)` so you know how many numbers MAX actually compared.

MIN Function in Excel

1. What is MIN?

The MIN function returns the smallest number in a set of values. It solves the problem of finding the lowest value in a long list, such as the lowest sale, the lowest score, or the cheapest price. A Data Analyst uses it to find the weakest results, the best (lowest) costs, and the starting point of a range of values.

2. Syntax

`=MIN(number1, [number2], ...)`

Argument	Required?	What it accepts
`number1`	Required	A number, a cell reference, or a range (for example `B2:B5`)
`number2, ...`	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

Most often you will use a single range, such as `=MIN(B2:B5)`. You can also mix a range with a number: `=MIN(B2:B5, 20000)` returns whichever is smaller, the lowest value in the range or 20000.

3. Simple Example

Find the lowest sales figure:

	A	B
1	Employee	Sales
2	Rahim	25000
3	Karim	42000
4	Hasan	35000
5	Nabil	50000
6	**Minimum**	**25000**

Formula in B6: `=MIN(B2:B5)`

Excel compares the four values in B2 to B5 (25000, 42000, 35000, 50000) and returns the smallest one. **Result: 25000**

4. Another Example

Find the lowest price. The Monitor price is not decided yet:

	A	B
1	Product	Price
2	Mouse	25
3	Keyboard	45
4	Monitor	TBD
5	Headset	18
6	Webcam	60
7	**Lowest**	**18**

Formula in B7: `=MIN(B2:B6)`

Excel ignores the text "TBD" and compares only 25, 45, 18, and 60. **Result: 18**

5. When to Use It

- Find the lowest sales figure in a period
- Find the lowest exam score to see who needs support
- Find the lowest salary in a department
- Find the cheapest price or lowest cost, such as the best supplier quote
- Find the earliest date in a list, such as the first order date (dates are stored as numbers)

6. Important Things to Remember

- Text and empty cells inside a range are ignored. They are not counted as zero.
- A cell containing 0 is counted, so a 0 entered for missing data becomes the minimum. Among negative numbers, the smallest is the one furthest from zero, so -20 is smaller than -5.
- If a referenced cell contains an error such as `#N/A`, the whole MIN result shows that error.
- Numbers stored as text (often left-aligned) are ignored in a range. Convert them to real numbers first.
- For dates, MIN returns the earliest date. Format the result cell as a date to read it.
- If the range has no numbers at all, MIN returns 0.

7. Common Mistakes

Mistake: Selecting an incomplete range, such as leaving out the last row. **Fix:** Check the highlighted range after typing the formula.

Mistake: Typing 0 for missing values, so MIN returns 0. **Fix:** Leave missing values blank, or use text such as "TBD", so MIN ignores them.

Mistake: Getting a wrong minimum because some numbers are stored as text. **Fix:** Convert them to numbers (for example with Text to Columns) and check that they right-align.

Mistake: Expecting MIN to show who has the lowest value. **Fix:** MIN returns only the number. Use a lookup to find the name (see the tip below).

8. Data Analyst Tip

MIN gives the value, not the name. To find who has it, use `=INDEX(A2:A5, MATCH(MIN(B2:B5), B2:B5, 0))`, which returns Rahim for the first example. MIN and MAX together show the spread of your data: `=MAX(B2:B5)-MIN(B2:B5)` gives 25000.

SEQUENCE Function in Excel

1. What is SEQUENCE?

The SEQUENCE function creates a list or grid of numbers automatically from a single formula. It solves the problem of typing or dragging numbers by hand, such as 1, 2, 3 or 1001, 1002, 1003. A Data Analyst uses it to create serial numbers, ID lists, and index numbers that update when the data changes.

2. Syntax

`=SEQUENCE(rows, [columns], [start], [step])`

Argument	Required?	What it accepts
`rows`	Required	How many rows of numbers to create (a whole number of 1 or more)
`columns`	Optional	How many columns to create. The default is 1.
`start`	Optional	The first number in the sequence. The default is 1.
`step`	Optional	The amount added each time. The default is 1. A negative step counts down.

Excel fills the grid row by row. `=SEQUENCE(2,3)` gives 1, 2, 3 in the first row and 4, 5, 6 in the second row. `=SEQUENCE(5,1,10,5)` gives 10, 15, 20, 25, 30 in one column.

3. Simple Example

Generate the numbers 1 to 5 in a single column:

	A
1	**Result**
2	**1**
3	**2**
4	**3**
5	**4**
6	**5**

Formula in A2 only: `=SEQUENCE(5)`

Excel creates 5 numbers, starting at 1 and adding 1 each time. It places them in A2 to A6 automatically. This is called "spilling".
Result: 1, 2, 3, 4, 5

4. Another Example

Create order numbers that start at 1001 for four customers:

	A	B
1	Customer	**Order ID**
2	Aisha	**1001**
3	Rahim	**1002**
4	Nabila	**1003**
5	Tanvir	**1004**

Formula in B2 only: `=SEQUENCE(4, 1, 1001)`

Excel creates 4 rows and 1 column, starting at 1001 and adding 1 each time. The step is left out, so it uses the default of 1. **Result: 1001 to 1004**

5. When to Use It

- Create serial numbers or ID numbers for a list of records
- Create month numbers 1 to 12 for a yearly report
- Build a list of dates by using a date as the start value (format the cells as dates)
- Create index or helper numbers for other formulas
- Generate quick sample numbers for practice or testing

6. Important Things to Remember

- SEQUENCE is a dynamic array function. One formula in one cell fills many cells. This is called "spilling".
- Numbers fill row by row: left to right across the first row, then down to the next row.
- The cells where the numbers will spill must be empty. If they are not, Excel shows `#SPILL!`.
- If you leave out `columns`, `start`, or `step`, Excel uses 1 for each one.
- `rows` and `columns` must be 1 or more. The `step` can be negative to count down, for example `=SEQUENCE(5,1,10,-2)` gives 10, 8, 6, 4, 2.
- SEQUENCE is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Getting `#SPILL!` because cells in the spill area already contain data. **Fix:** Clear those cells, or move the formula to an empty area.

Mistake: Trying to edit one of the spilled numbers. **Fix:** Only the first cell holds the formula. Edit that cell, and the whole result updates.

Mistake: Mixing up rows and columns, such as `=SEQUENCE(1,5)` (one row) when a column was needed. **Fix:** Remember the order: rows first, then columns.

Mistake: Trying to set a step without a start, such as `=SEQUENCE(5,1,2)`, which starts at 2 with a step of 1. **Fix:** Fill the arguments in order: `=SEQUENCE(5,1,1,2)`.

8. Data Analyst Tip

Use SEQUENCE for numbering that adjusts itself. For example, `=SEQUENCE(COUNTA(A2:A100))` creates one number for each filled cell in A2:A100, so the numbering grows or shrinks with your data. This is much safer than typing numbers by hand or dragging them down.

SORT Function in Excel

1. What is SORT?

The SORT function returns a sorted copy of a list or table, in ascending or descending order. It solves the problem of putting data in a useful order, such as names from A to Z or sales from highest to lowest. A Data Analyst uses it to build ranked lists and ordered reports that update automatically when the source data changes, without changing the original table.

2. Syntax

`=SORT(array, [sort_index], [sort_order], [by_col])`

Argument	Required?	What it accepts
`array`	Required	The range or table to sort (for example `A2:C7`)
`sort_index`	Optional	The number of the column to sort by, counted within the array. The default is 1 (the first column).
`sort_order`	Optional	`1` for ascending, smallest to largest or A to Z (default). `-1` for descending.
`by_col`	Optional	`FALSE` (default) sorts rows. `TRUE` sorts columns from left to right. Beginners rarely need TRUE.

Text sorts alphabetically and numbers sort from smallest to largest. In `A2:C7`, a `sort_index` of 3 means the third column of the array, which is column C.

3. Simple Example

Sort a list of products from A to Z:

	A	B	C
1	Product		**Sorted Product**
2	Orange		**Apple**
3	Apple		**Banana**
4	Mango		**Grape**
5	Banana		**Kiwi**
6	Grape		**Mango**
7	Kiwi		**Orange**

Formula in C2 only: `=SORT(A2:A7, 1, 1)` (the heading in C1 is typed)

The list has one column, so the sort index is 1. The sort order 1 means ascending. Excel sorts the six products from A to Z and spills them into C2 to C7. **Result: Apple, Banana, Grape, Kiwi, Mango, Orange**

4. Another Example

Rank employees by sales, highest first:

	A	B	C	D	E	F	G
1	Name	Dept	Sales		**Name**	**Dept**	**Sales**
2	Ayesha	IT	850		**Nabila**	**IT**	**910**
3	Rahim	HR	620		**Ayesha**	**IT**	**850**
4	Nabila	IT	910		**Fahim**	**Finance**	**780**
5	Fahim	Finance	780		**Sadia**	**IT**	**660**
6	Sadia	IT	660		**Rahim**	**HR**	**620**
7	Rafi	Sales	540		**Rafi**	**Sales**	**540**

Formula in E2 only: `=SORT(A2:C7, 3, -1)` (the headings in E1:G1 are typed)

Excel sorts the whole table by its 3rd column (Sales). The sort order -1 means descending, so the highest sales come first. Each name stays with its own department and sales. **Result: Nabila at the top, Rafi at the bottom**

5. When to Use It

- Rank top performers, products, or regions by sales or score
- Create alphabetical lists of names, products, or customers
- Keep a report in order automatically as the data changes
- Sort the results of other functions, such as FILTER or UNIQUE
- Show a sorted view of the data while keeping the original table untouched

6. Important Things to Remember

- SORT is a dynamic array function. One formula fills many cells, so the cells for the result must be empty, or Excel shows `#SPILL!`.
- The original data is not changed. SORT returns a sorted copy and updates when the source changes.
- `sort_index` is the position of the column inside the array, not the column letter on the sheet.
- `sort_order` must be `1` (ascending) or `-1` (descending). Any other number gives `#VALUE!`.
- Select the whole table in `array` so that all columns move together with the sorted column.
- SORT is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Selecting only the column to be sorted, so the other columns do not follow it. **Fix:** Select the whole table in `array` and use `sort_index` to choose the column.

Mistake: Using the sheet column number for `sort_index`. For `B2:D7`, a `sort_index` of 3 sorts by column D, not column C. **Fix:** Count from the first column of the array.

Mistake: Including the header row in the array, so the heading is sorted into the data. **Fix:** Start the range at the first data row.

Mistake: Using 0 or 2 for `sort_order`, which gives `#VALUE!`. **Fix:** Use `1` for ascending or `-1` for descending.

8. Data Analyst Tip

To sort by two columns, give array constants for the second and third arguments. For example, `=SORT(A2:C7, {2,3}, {1,-1})` sorts by Department A to Z, then by Sales high to low within each department. You can also wrap SORT around other functions, such as `=SORT(UNIQUE(B2:B7))`, for a sorted list of unique values.

TRIM Function in Excel

1. What is TRIM?

The TRIM function removes extra spaces from text: spaces at the start, spaces at the end, and repeated spaces between words. It leaves a single space between words. It solves the problem of messy text, often from imported or copied data, where hidden spaces cause names or categories to look the same but not match. A Data Analyst uses it as an early step when cleaning data.

2. Syntax

`=TRIM(text)`

Argument	Required?	What it accepts
`text`	Required	The text to clean: a cell reference (for example `B2`) or text typed in quotes (for example `" Hello "`)

TRIM has only one argument. The result is a cleaned copy of the text; the original cell does not change.

3. Simple Example

Clean email addresses with extra spaces (quotation marks show the spaces):

	A	B	C
1	Customer	Email	**Clean Email**
2	C001	" aisha@mail.com"	**"aisha@mail.com"**
3	C002	"rahim@mail.com "	**"rahim@mail.com"**
4	C003	" nabila@mail.com "	**"nabila@mail.com"**
5	C004	"tanvir@mail.com "	**"tanvir@mail.com"**

Formula in C2: `=TRIM(B2)`, then copied down to C5.

For row 2, Excel removes the two spaces before the address. For row 3, it removes the two spaces after it. The email text itself does not change. **Result: clean addresses with no leading or trailing spaces**

4. Another Example

Clean city names in an imported sales file:

	A	B	C
1	Order	City	**Clean City**
2	O101	"Dhaka "	**"Dhaka"**
3	O102	" Chattogram"	**"Chattogram"**
4	O103	"Cox's Bazar"	**"Cox's Bazar"**
5	O104	"Sylhet "	**"Sylhet"**

Formula in C2: `=TRIM(B2)`, then copied down to C5.

"Dhaka " with a trailing space looks the same as "Dhaka", but Excel sees them as different values, so counts and groupings by city would be split. In row 4, TRIM reduces the three spaces inside "Cox's Bazar" to one. **Result: values that match exactly**

5. When to Use It

- Clean data imported from other systems, websites, or files
- Clean customer names and email addresses before analysis
- Fix lookups, counts, and filters that fail because of hidden spaces
- Standardize product names and categories before grouping or building a pivot table
- Prepare text columns before joining or comparing them

6. Important Things to Remember

- TRIM removes leading spaces, trailing spaces, and extra spaces between words. It keeps one single space between words.
- TRIM does not change the letters, capitalization, or order of the text.
- The original cell stays unchanged. Put the formula in a new column.
- The result is text. A number with spaces around it becomes a text value after TRIM.
- TRIM removes only the normal space character. It does not remove non-breaking spaces (common in web data) or line breaks.
- TRIM never removes the single space between two words, so it cannot turn "Aisha Khan" into "AishaKhan".

7. Common Mistakes

Mistake: Expecting the original column to be cleaned. **Fix:** TRIM works in a new column. Copy the results and paste them over the original as values if you want to replace it.

Mistake: Deleting the original column while the TRIM formulas still depend on it, which gives `#REF!`. **Fix:** Paste the cleaned results as values first, then delete the original.

Mistake: Expecting TRIM to remove every space. **Fix:** To remove all spaces, including those between words, use `SUBSTITUTE(B2, " ", "")`.

Mistake: Thinking TRIM failed because the text looks unchanged. **Fix:** The data may contain non-breaking spaces. Use `TRIM(SUBSTITUTE(B2, CHAR(160), " "))`.

8. Data Analyst Tip

Before using lookups, counts, or pivot tables, check key columns such as IDs, names, and categories for hidden spaces. Type `=LEN(B2)-LEN(TRIM(B2))` in a helper column. Any result above 0 shows how many extra spaces that cell contains, so you know which rows need cleaning.

UNIQUE Function in Excel

1. What is UNIQUE?

The UNIQUE function takes a list and returns each different value only once, removing the repeats. It solves the problem of long columns where the same item appears many times, such as product categories, regions, or customers. A Data Analyst uses it to get a clean list of distinct items without changing the original data.

2. Syntax

`=UNIQUE(array, [by_col], [exactly_once])`

Argument	Required?	What it accepts
`array`	Required	The range to check for repeats (for example `A2:A9`)
`by_col`	Optional	`FALSE` (default) compares rows. `TRUE` compares columns. Most beginners leave this out.
`exactly_once`	Optional	`FALSE` (default) returns every different value once. `TRUE` returns only values that appear exactly one time.

If the range has two or more columns, UNIQUE compares whole rows (see Example 4). With TRUE as the third argument, `=UNIQUE(A2:A9, , TRUE)` returns only the values that appear exactly once.

3. Simple Example

List the unique product categories:

	A	B	C
1	Category		**Unique Category**
2	Electronics		**Electronics**
3	Furniture		**Furniture**
4	Electronics		**Clothing**
5	Clothing		**Books**
6	Furniture		
7	Books		
8	Clothing		
9	Electronics		

Formula in C2 only: `=UNIQUE(A2:A9)`

Excel goes down A2:A9 and keeps each category the first time it appears: Electronics, Furniture, Clothing, Books. The 8 entries become 4 unique values, spilled into C2 to C5. **Result: 4 unique categories**

4. Another Example

Remove repeated rows when two columns are checked together:

	A	B	C	D	E
1	Product	Region		**Product**	**Region**
2	Laptop	Dhaka		**Laptop**	**Dhaka**
3	Phone	Chattogram		**Phone**	**Chattogram**
4	Laptop	Dhaka		**Mouse**	**Dhaka**
5	Mouse	Dhaka		**Laptop**	**Sylhet**
6	Phone	Chattogram			
7	Laptop	Sylhet			

Formula in D2 only: `=UNIQUE(A2:B7)` (the headings in D1:E1 are typed)

Excel compares whole rows. Rows 2 and 4 match (Laptop, Dhaka) and rows 3 and 6 match (Phone, Chattogram), so each is kept once. Laptop, Sylhet is a different pair. **Result: 4 unique rows**

5. When to Use It

- Get a list of unique products, categories, or regions from a long column
- Remove duplicates from a list without changing the original data
- Find unique combinations, such as each Product and Region pair
- Find customers or items that appear only once
- Create a list that feeds a summary table

6. Important Things to Remember

- UNIQUE is a dynamic array function. You type one formula and the results spill down into the cells below.
- Results appear in the order each value first appears. The original data is not changed.
- UNIQUE is not case-sensitive ("dhaka" and "Dhaka" are the same), but extra spaces make values different.
- With two or more columns, UNIQUE compares whole rows, not single columns.
- Blank cells inside the selected range appear in the result as 0, so select only your data.
- UNIQUE is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Selecting extra empty cells, such as `A2:A12` when data ends at A9, so a 0 appears in the result. **Fix:** Select only the rows that contain data.

Mistake: Including the header in the range, so the header appears as one of the unique values. **Fix:** Start the range from the first data row.

Mistake: Seeing "duplicates" in the result, such as "Dhaka" and "Dhaka ". **Fix:** Hidden spaces cause this. Clean the data with TRIM.

Mistake: Getting `#SPILL!` because cells below the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

Combine UNIQUE with other functions. `=COUNTA(UNIQUE(A2:A9))` counts how many different categories there are, which is 4 in the first example. `=SORT(UNIQUE(A2:A9))` gives the unique list in alphabetical order, which suits summary tables.

FILTER Function in Excel

1. What is FILTER?

The FILTER function returns only the rows of a table that meet a condition you set. It solves the problem of pulling out a subset of data, such as the employees in one department or the orders from one region. A Data Analyst uses it to build reports and lists that update automatically when the source data changes, without touching the original table.

2. Syntax

`=FILTER(array, include, [if_empty])`

Argument	Required?	What it accepts
`array`	Required	The range or table to return data from (for example `A2:C7`)
`include`	Required	A condition that is TRUE or FALSE for each row, such as `B2:B7="IT"`. It must have the same number of rows as `array`.
`if_empty`	Optional	What to show if no rows match, such as `"No data"`. If it is left out, Excel shows `#CALC!`.

For several conditions, put each one in brackets and join them: `*` means AND (both must be true) and `+` means OR (at least one must be true).

3. Simple Example

Show only the students from the IT department:

	A	B	C	D	E	F	G
1	Name	Dept	Score		**Name**	**Dept**	**Score**
2	Ayesha	IT	85		**Ayesha**	**IT**	**85**
3	Rahim	HR	78		**Nabila**	**IT**	**92**
4	Nabila	IT	92		**Sadia**	**IT**	**76**
5	Fahim	Finance	88				
6	Sadia	IT	76				
7	Rafi	Sales	81				

Formula in E2 only: `=FILTER(A2:C7, B2:B7="IT", "No data")` (the headings in E1:G1 are typed)

Excel checks each row of B2:B7 and keeps the rows where the department is "IT". Rows 2, 4, and 6 match, and all three columns of those rows are returned. **Result: 3 rows**

4. Another Example

Show orders from the North region with an amount over 300:

	A	B	C	D	E	F	G
1	Order	Region	Amount		**Order**	**Region**	**Amount**
2	O101	North	250		**O103**	**North**	**520**
3	O102	South	400		**O105**	**North**	**480**
4	O103	North	520				
5	O104	East	350				
6	O105	North	480				
7	O106	South	180				

Formula in E2 only: `=FILTER(A2:C7, (B2:B7="North")*(C2:C7>300), "No orders")`

Each condition gives TRUE or FALSE for every row. Multiplying them keeps only rows where both are TRUE: O103 and O105. O101 is North but only 250. O102 is over 300 but South. **Result: 2 rows**

5. When to Use It

- Pull all records for one department, region, or status
- List top performers, such as scores above a target
- Create reports that update automatically when the data changes
- Extract orders that meet several conditions, such as region and amount
- Exclude blank rows from a list

6. Important Things to Remember

- FILTER is a dynamic array function. One formula returns many cells, so the cells for the result must be empty, or Excel shows `#SPILL!`.
- The `include` condition must have the same number of rows as `array`.
- If no rows match and `if_empty` is left out, Excel shows `#CALC!`. Always add a message such as `"No data"`.
- Put each condition in brackets when joining them with `*` (AND) or `+` (OR).
- FILTER does not change or hide the original data, and the result updates when the source data changes.
- FILTER is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Using ranges of different sizes, such as `A2:C7` with `B2:B8`, which gives `#VALUE!`. **Fix:** Select the same rows for `array` and `include`.

Mistake: Forgetting the brackets around each condition, such as `B2:B7="North"*C2:C7>300`, which gives wrong results. **Fix:** Write `(B2:B7="North")*(C2:C7>300)`.

Mistake: Forgetting quotation marks around text, such as `B2:B7=IT`. **Fix:** Write the text in quotes: `B2:B7="IT"`.

Mistake: Seeing `#CALC!` when no rows match. **Fix:** Add the third argument, for example `"No data"`.

8. Data Analyst Tip

Type the condition value in a cell and refer to it, such as `=FILTER(A2:C7, B2:B7=F1, "No data")`, so you can change the department in F1 and the list updates. Combine FILTER with SORT to rank the results, for example `=SORT(FILTER(A2:C7, B2:B7=F1), 3, -1)` lists the department by score, highest first.

VLOOKUP in Excel

1. What is VLOOKUP?

VLOOKUP stands for Vertical Lookup. It searches for a value in the first column of a table and returns a value from another column in the same row. It solves the problem of finding related information, such as the salary of an employee or the price of a product, using an ID. A Data Analyst uses it to pull data from one table into another without searching by hand.

2. Syntax

`=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

Argument	Required?	What it accepts
`lookup_value`	Required	The value to find, such as a number, text in quotes, or a cell reference
`table_array`	Required	The table to search (for example `A2:D4`). The lookup value must be in its first column.
`col_index_num`	Required	The number of the column to return, counted inside the table. The first column is 1.
`range_lookup`	Optional	`FALSE` for an exact match. `TRUE` for an approximate match (the default if you leave it out).

Use `FALSE` whenever you look up IDs, names, or codes. Use `TRUE` only for ranges, such as commission bands or grades, and only when the first column is sorted from smallest to largest.

3. Simple Example

Find the salary for employee ID 103 (exact match):

	A	B	C	D	E	F	G
1	ID	Name	Dept	Salary		ID	**Salary**
2	101	Rahim	Sales	30000		103	**40000**
3	102	Karim	HR	35000			
4	103	Hasan	IT	40000			

Formula in G2: `=VLOOKUP(F2, A2:D4, 4, FALSE)`

Excel finds 103 in the first column, A2:A4 (row 4). Then it returns the value from column 4 of the table, Salary, in that row. **Result: 40000**

4. Another Example

Find the commission rate for sales of 12000 (approximate match):

	A	B	C	D	E
1	Sales From	Rate		Sales	**Rate**
2	0	0%		12000	**5%**
3	5000	2%			
4	10000	5%			
5	20000	8%			

Formula in E2: `=VLOOKUP(D2, A2:B5, 2, TRUE)`

12000 is not in the list. With TRUE, Excel finds the largest value that is not more than 12000, which is 10000, and returns its rate. **Result: 5%**

5. When to Use It

- Find an employee's salary or department using an employee ID
- Retrieve a product name or price using a product ID
- Match student IDs to marks or grades
- Apply commission rates, tax slabs, or grade bands (approximate match)
- Bring customer details into an orders table

6. Important Things to Remember

- The lookup value must be in the first column of `table_array`. VLOOKUP only returns columns to the right of it.
- `col_index_num` counts columns inside the table, not sheet column letters. In `B2:E10`, column 1 is B.
- If `range_lookup` is left out, Excel uses `TRUE` (approximate), which can give wrong results for IDs. Always type `FALSE` for exact matches.
- With `TRUE`, the first column must be sorted smallest to largest. Excel returns the largest value that is not more than the lookup value.
- If the value is not found, the result is `#N/A`. If `col_index_num` is larger than the table, the result is `#REF!`.
- If the first column has duplicates, VLOOKUP returns the first match. Matching is not case-sensitive.

7. Common Mistakes

Mistake: Leaving out `FALSE`, so Excel uses approximate matching and returns the wrong row. **Fix:** Always end an exact lookup with `, FALSE)`.

Mistake: Copying the formula down without locking the table, so the range moves and gives `#N/A`. **Fix:** Use `$` signs: `$A$2:$D$10`.

Mistake: Having the lookup value outside the first column of the table. **Fix:** Start the table at the column that holds the lookup value, or use XLOOKUP or INDEX + MATCH.

Mistake: Getting `#N/A` for an ID that looks correct, because one is stored as text or has extra spaces. **Fix:** Make both sides the same type and clean the data with TRIM.

8. Data Analyst Tip

Put the lookup value in a cell, lock the table with $, and wrap the formula in IFERROR so missing IDs show a clear message: `=IFERROR(VLOOKUP(F2, $A$2:$D$4, 4, FALSE), "Not Found")`. In Excel 2021 or Microsoft 365, XLOOKUP is easier for new work because it needs no column number and can look to the left.

XLOOKUP Function in Excel

1. What is XLOOKUP?

The XLOOKUP function searches for a value in one column (or row) and returns the matching value from another column. It solves the problem of finding related information, such as the department of an employee or the price of a product, using an ID. A Data Analyst uses it to pull data from one table into another without searching manually.

2. Syntax

`=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

Argument	Required?	What it accepts
`lookup_value`	Required	The value to find (for example `"E103"` or a cell like `A6`)
`lookup_array`	Required	The single column or row to search in (for example `A2:A5`)
`return_array`	Required	The column or row that holds the result (for example `C2:C5`). It must be the same size as `lookup_array`.
`if_not_found`	Optional	What to show if no match exists, such as `"Not Found"`. Without it, Excel shows `#N/A`.
`match_mode`	Optional	`0` exact match (default), `-1` exact or next smaller, `1` exact or next larger, `2` wildcard
`search_mode`	Optional	`1` search first to last (default), `-1` search last to first

Beginners usually need only the first three arguments, and the fourth when missing values are possible.

3. Simple Example

Find the department of Employee E103 (typed in A6):

	A	B	C
1	Employee ID	Employee Name	Department
2	E101	Aisha	Marketing
3	E102	Rahim	Finance
4	E103	Nabila	Analytics
5	E104	Tanvir	Operations
6	**E103**	**Result**	**Analytics**

Formula in C6: `=XLOOKUP(A6, A2:A5, C2:C5)`

Excel finds E103 in A2:A5 (it is in row 4), then returns the value from the same row in C2:C5. **Result: Analytics**. You can also type the ID directly: `=XLOOKUP("E103", A2:A5, C2:C5)`.

4. Another Example

Find the price of Product P203, with a message if the ID is missing:

	A	B	C
1	Product ID	Product	Price
2	P201	Notebook	120
3	P202	Pen	25
4	P203	Stapler	350
5	P204	Marker	90
6	**P203**	**Result**	**350**

Formula in C6: `=XLOOKUP(A6, A2:A5, C2:C5, "Not Found")`

Excel finds P203 in A2:A5 and returns the price from the same row. **Result: 350**. If A6 contained P999, which is not in the list, Excel would show "Not Found" instead of `#N/A`.

5. When to Use It

- Find employee details, such as department, using an employee ID
- Retrieve product prices or names using a product ID
- Pull customer information, such as city or segment, using a customer ID
- Match sales or orders to details stored in another table
- Check which IDs from one list are missing from another list

6. Important Things to Remember

- XLOOKUP finds an exact match by default. You do not need an extra setting for it.
- `lookup_array` and `return_array` must be the same size, and each should be one column (or one row).
- The result column can be to the left or right of the lookup column. There are no column numbers to count.
- If there are duplicate values in the lookup column, XLOOKUP returns the first match.
- If nothing matches and `if_not_found` is not used, the result is `#N/A`. Matching is not case-sensitive.
- XLOOKUP is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Using arrays of different sizes, such as `A2:A5` with `C2:C6`. **Fix:** Select the same rows for both.

Mistake: Selecting the whole table as the lookup or return array. **Fix:** Select only one column for each array.

Mistake: Typing a text lookup value without quotation marks, such as `E103`. **Fix:** Write `"E103"`, or put the value in a cell and refer to that cell.

Mistake: Getting `#N/A` for an ID that looks correct, because of extra spaces or numbers stored as text. **Fix:** Clean the data so both lists use the same format.

8. Data Analyst Tip

When you bring data from one table into another, use the `if_not_found` argument with a clear message such as "Not Found", then filter that column. This quickly shows IDs that are missing from the source table. Put the lookup value in a cell and lock the arrays with $, for example `$A$2:$A$5`, so you can copy the formula down.

INDEX + MATCH in Excel

1. What is INDEX + MATCH?

INDEX and MATCH are two functions used together to look up a value in a table. MATCH finds the position of an item in a list, and INDEX returns the value at that position from another column. Together they solve the problem of finding related information, such as the sales amount for a product. A Data Analyst uses them because they are more flexible than VLOOKUP: they can return a value from a column to the left of the search column, and they do not break when columns are inserted.

2. Syntax

`=INDEX(return_range, MATCH(lookup_value, lookup_array, 0))`

Argument	Required?	What it accepts
`return_range`	Required	The column (or row) that holds the value you want back (for example `C2:C7`)
`lookup_value`	Required	The value to find: text in quotes, a number, or a cell reference (for example `B8`)
`lookup_array`	Required	The column (or row) to search in (for example `B2:B7`). It must be the same size as `return_range`.
`match_type`	Optional*	`0` exact match, `1` approximate (data sorted ascending), `-1` approximate (data sorted descending)

*Always type `0` for an exact match. If it is left out, MATCH uses `1` and can return wrong results. MATCH(lookup_value, lookup_array, match_type) returns a position number. INDEX(array, row_num) returns the value at that position.

3. Simple Example

Find the sales amount for the product "Monitor" (typed in B8):

	A	B	C
1	Product ID	Product Name	Sales Amount
2	P001	Laptop	1200
3	P002	Mouse	150
4	P003	Keyboard	300
5	P004	Monitor	800
6	P005	Headphones	250
7	P006	Webcam	180
8	**Result**	**Monitor**	**800**

Formula in C8: `=INDEX(C2:C7, MATCH(B8, B2:B7, 0))`

Step 1: MATCH finds "Monitor" in B2:B7. It is the 4th item, so MATCH returns 4. Step 2: INDEX returns the 4th value in C2:C7. **Result: 800**

4. Another Example

Find the name for employee ID E103. The name is to the left of the ID:

	A	B
1	Name	Employee ID
2	Aisha	E101
3	Rahim	E102
4	Nabila	E103
5	Tanvir	E104
6	**Nabila**	**E103**

Type the ID in B6. Formula in A6: `=INDEX(A2:A5, MATCH(B6, B2:B5, 0))`

MATCH finds E103 in B2:B5, in position 3. INDEX returns the 3rd value in A2:A5. **Result: Nabila**. VLOOKUP cannot return a column to the left of the search column, but INDEX + MATCH can.

5. When to Use It

- Find a sales amount, price, or other detail using an ID or name
- Return a value from a column to the left of the search column
- Pull specific data into reports and dashboards
- Build lookups that keep working when columns are inserted or moved
- Replace VLOOKUP in large tables

6. Important Things to Remember

- MATCH finds the position of the value in its range, not the sheet row number. INDEX uses that position in its own range.
- Use `0` as the last MATCH argument for an exact match. This is what you need in most cases.
- `return_range` and `lookup_array` must be the same size and start on the same row.
- Match types `1` and `-1` need sorted data (ascending for `1`, descending for `-1`).
- If MATCH finds no match, the result is `#N/A`. Matching is not case-sensitive.
- Text typed in the formula goes in quotation marks, or use a cell reference instead.

7. Common Mistakes

Mistake: Leaving out the `0`, so MATCH uses approximate matching and returns wrong values. **Fix:** Always write `, 0)` at the end of MATCH.

Mistake: Searching in the wrong column, such as looking for "Monitor" in the column of product IDs, which gives `#N/A`. **Fix:** `lookup_array` must be the column that contains the value you are looking for.

Mistake: Ranges that do not line up, such as `C2:C7` with `B3:B8`. **Fix:** Select the same rows in both ranges.

Mistake: Getting `#N/A` for a value that looks correct, because of extra spaces or different formats. **Fix:** Clean the data (for example with TRIM) and check that both sides are text or both are numbers.

8. Data Analyst Tip

Put the lookup value in a cell and lock the ranges with $, such as `$B$2:$B$7`, so you can copy the formula down. Wrap the formula in IFERROR, for example `=IFERROR(INDEX(...), "Not Found")`, so missing items show a clear message instead of `#N/A`.

INDEX + MATCH + MATCH in Excel

1. What is INDEX + MATCH + MATCH?

This is a two-way lookup. It finds the value where a row and a column cross in a table. One MATCH finds the row position, a second MATCH finds the column position, and INDEX returns the value at that crossing. It solves the problem of finding a number in a grid, such as the sales of one product in one region. A Data Analyst uses it to build flexible lookups that work even when rows or columns are added or moved.

2. Syntax

`=INDEX(return_range, MATCH(row_value, row_range, 0), MATCH(column_value, column_range, 0))`

Argument	Required?	What it accepts
`return_range`	Required	The table body only, without headings or row labels (for example `B2:D5`)
`row_value`	Required	The row label to find, such as `"East"` or a cell reference
`row_range`	Required	The column of row labels (for example `A2:A5`). It must have the same number of rows as `return_range`.
`column_value`	Required	The column heading to find, such as `"Laptop"` or a cell reference
`column_range`	Required	The row of headings (for example `B1:D1`). It must have the same number of columns as `return_range`.

The first MATCH gives the row number and the second gives the column number, always in that order. The `0` at the end of each MATCH means exact match. Always include it.

3. Simple Example

Find the Laptop sales for the East region:

	A	B	C	D	E	F	G	H
1	Region	Laptop	Mouse	Keyboard		Region	Product	**Sales**
2	North	1200	150	300		East	Laptop	**1000**
3	South	800	120	250				
4	East	1000	140	280				
5	West	900	110	220				

Formula in H2: `=INDEX(B2:D5, MATCH(F2, A2:A5, 0), MATCH(G2, B1:D1, 0))`

The first MATCH finds East in A2:A5: position 3. The second MATCH finds Laptop in B1:D1: position 1. INDEX returns the value at row 3, column 1 of B2:D5. **Result: 1000**

4. Another Example

Find the sales of one employee in one month:

	A	B	C	D	E	F	G	H
1	Name	Jan	Feb	Mar		Name	Month	**Sales**
2	Ayesha	5000	5200	5100		Tanvir	Feb	**3900**
3	Rahim	4200	4500	4400				
4	Nabila	6100	6000	6300				
5	Tanvir	3800	3900	4000				

Formula in H2: `=INDEX(B2:D5, MATCH(F2, A2:A5, 0), MATCH(G2, B1:D1, 0))`

The first MATCH finds Tanvir in A2:A5: position 4. The second MATCH finds Feb in B1:D1: position 2. INDEX returns the value at row 4, column 2 of B2:D5. **Result: 3900**

5. When to Use It

- Find sales by region and product, or by employee and month
- Look up a price from a price list using item and size
- Find stock levels by item and warehouse
- Find a score by student and subject
- Build interactive reports where the user changes the row and column values

6. Important Things to Remember

- The first MATCH finds the row and the second MATCH finds the column. The order inside INDEX is row, then column.
- `return_range` holds only the data. Do not include the headings or row labels.
- `row_range` and `column_range` must line up with `return_range`, with the same number of rows and columns.
- Use `0` in both MATCH functions for an exact match. If it is left out, MATCH uses `1` and can return wrong results.
- If a label is not found, the result is `#N/A`. Wrap the formula in IFERROR for a message.
- This formula works in every version of Excel, and labels are not case-sensitive.

7. Common Mistakes

Mistake: Swapping the two MATCH functions, which gives a wrong value or an error. **Fix:** Put the row MATCH first and the column MATCH second.

Mistake: Including the headings in `return_range`, which shifts every position by one. **Fix:** Start `return_range` at the first data cell.

Mistake: Ranges that do not line up, such as `A2:A5` with `B2:D6`. **Fix:** Use the same rows for `row_range` and `return_range`.

Mistake: Leaving out the `0`, so MATCH looks for an approximate match. **Fix:** Always end each MATCH with `, 0)`.

8. Data Analyst Tip

When rows are identified by two labels, such as Region and Product with quarters across the top, match both with `MATCH(1, (A2:A7="East")*(B2:B7="Laptop"), 0)`. The full formula is `=INDEX(C2:F7, MATCH(1, (A2:A7="East")*(B2:B7="Laptop"), 0), MATCH("Q2", C1:F1, 0))`. In Excel 2019 or earlier, confirm it with Ctrl + Shift + Enter.

TOCOL Function in Excel

1. What is TOCOL?

The TOCOL function takes a range with many rows and columns and stacks all its values into a single column. It solves the problem of data that is spread across a grid when you need one simple list. A Data Analyst uses it to flatten tables so the values can be sorted, counted, filtered, or passed to other functions.

2. Syntax

`=TOCOL(array, [ignore], [scan_by_column])`

Argument	Required?	What it accepts
`array`	Required	The range or array to convert into one column (for example `A1:C3`)
`ignore`	Optional	What to skip: `0` keep all (default), `1` ignore blanks, `2` ignore errors, `3` ignore blanks and errors
`scan_by_column`	Optional	`FALSE` (default) reads row by row, left to right. `TRUE` reads column by column, top to bottom.

To use `scan_by_column` and keep the default for `ignore`, leave the middle argument empty or use 0: `=TOCOL(A1:C3, , TRUE)`.

3. Simple Example

Turn a 3 × 3 grid into one column, ignoring blank cells:

	A	B	C		E
1	Apple	Banana	Orange		**Apple**
2	Mango		Grape		**Banana**
3	Kiwi	Pineapple			**Orange**
4					**Mango**
5					**Grape**
6					**Kiwi**
7					**Pineapple**

Formula in E1 only: `=TOCOL(A1:C3, 1)`

Excel reads row 1 (Apple, Banana, Orange), then row 2, then row 3, skipping the two blank cells (B2 and C3). The 7 values spill down E1 to E7. **Result: 7 values in one column**

4. Another Example

Stack monthly sales into one column, reading column by column:

	A	B	C	D		F
1	Region	Jan	Feb	Mar		**Result**
2	North	500	700	650		**500**
3	South	400	550	600		**400**
4						**700**
5						**550**
6						**650**
7						**600**

Formula in F2 only: `=TOCOL(B2:D3, , TRUE)`

With TRUE, Excel reads down the Jan column first (500, 400), then Feb (700, 550), then Mar (650, 600). With the default FALSE it would read row by row: 500, 700, 650, 400, 550, 600.

5. When to Use It

- Flatten a multi-column table into one list for analysis
- Combine several columns of names, IDs, or codes into one list
- Remove blanks and errors while reshaping data
- Prepare a list for functions such as UNIQUE, SORT, or COUNTIF
- Prepare wide data for charts or pivot tables

6. Important Things to Remember

- TOCOL is a dynamic array function. You type one formula and the results spill down into the cells below.
- By default, TOCOL reads row by row (`scan_by_column` is FALSE). Set it to TRUE to read column by column.
- By default, nothing is ignored, so blank cells appear in the result as 0 and errors are kept.
- Use `1` to skip blanks, `2` to skip errors, and `3` to skip both.
- The cells below the formula must be empty, or Excel shows `#SPILL!`.
- TOCOL is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Expecting blank cells to be skipped automatically, then finding 0 values in the result. **Fix:** Add the ignore argument: `=TOCOL(A1:C3, 1)`.

Mistake: Using `1` to remove blanks and errors, but still seeing `#N/A`. **Fix:** Use `3` to ignore both blanks and errors, or `2` for errors only.

Mistake: Getting the wrong order because of the reading direction. **Fix:** Remember that row by row is the default. Add `TRUE` as the third argument for column by column.

Mistake: Getting `#SPILL!` because cells below the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

Combine TOCOL with other functions to build clean lists. For example, `=SORT(UNIQUE(TOCOL(A1:C10, 3)))` flattens a table, removes blanks and errors, removes duplicates, and sorts the result in one formula.

TOCOL + UNIQUE in Excel

1. What is TOCOL + UNIQUE?

TOCOL and UNIQUE are two functions used together. TOCOL flattens a range with many rows and columns into a single column, and UNIQUE then removes the repeated values. Together they solve the problem of finding every different value that is spread across several columns, such as skills, tags, or product names. A Data Analyst uses them to build a clean list of distinct items from a grid of data in one formula.

2. Syntax

`=UNIQUE(TOCOL(array, [ignore], [scan_by_column]))`

Argument	Required?	What it accepts
`array`	Required	The range to flatten and check for repeats (for example `A2:C5`)
`ignore`	Optional	What TOCOL skips: `0` keep all (default), `1` ignore blanks, `2` ignore errors, `3` ignore blanks and errors
`scan_by_column`	Optional	`FALSE` (default) reads row by row. `TRUE` reads column by column. It only changes the order of the results.

Excel works from the inside out. TOCOL runs first and turns the range into one column. UNIQUE then keeps each value once, in the order it first appears.

3. Simple Example

Get the different fruits from a 3 × 3 grid:

	A	B	C	D	E
1	Apple	Banana	Apple		**Apple**
2	Orange	Grape	Banana		**Banana**
3	Mango	Orange	Kiwi		**Orange**
4					**Grape**
5					**Mango**
6					**Kiwi**

Formula in E1 only: `=UNIQUE(TOCOL(A1:C3, 1))`

TOCOL reads the 9 cells row by row into one column. UNIQUE removes the repeats of Apple, Banana, and Orange. **Result: 6 different fruits**

4. Another Example

List every skill in a grid that has blank cells:

	A	B	C	D	E
1	Skill 1	Skill 2	Skill 3		**Unique Skills**
2	Excel	SQL			**Excel**
3	SQL	Python	Excel		**SQL**
4	Excel				**Python**
5	Power BI	SQL	Python		**Power BI**

Formula in E2 only: `=UNIQUE(TOCOL(A2:C5, 1))` (the heading in E1 is typed)

The 1 in TOCOL skips the blank cells. Without it, the blanks would appear in the result as 0. **Result: Excel, SQL, Python, Power BI**

5. When to Use It

- Get a list of different items that are spread across several columns
- Build a clean list of skills, tags, or categories from a grid
- Create a list for a dropdown or a summary table
- Count how many different items exist across multiple columns
- Check which values appear anywhere in a multi-column range

6. Important Things to Remember

- Read the formula from the inside out: TOCOL flattens first, then UNIQUE removes repeats.
- By default, TOCOL keeps blank cells as 0, so a 0 can appear in the result. Use `1` to ignore blanks.
- Results appear in the order each value first appears. `scan_by_column` changes the reading direction and so the order.
- UNIQUE is not case-sensitive, but extra spaces make values different, so "Excel " and "Excel" count as separate values.
- The result spills into the cells below. They must be empty, or Excel shows `#SPILL!`.
- TOCOL is available in Microsoft 365 and Excel 2024, so this combination does not work in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Leaving out the ignore value, so a 0 appears in the list for every blank cell. **Fix:** Use `TOCOL(A2:C5, 1)`.

Mistake: Flattening more columns than needed, such as TOCOL around a FILTER result with several columns. Values from every column appear in the list. **Fix:** Filter only the column you need: `=UNIQUE(FILTER(A2:A7, B2:B7="Electronics"))`.

Mistake: Putting the TOCOL settings in the wrong brackets, such as `=UNIQUE(TOCOL(A2:C5), 1)`. **Fix:** Keep them inside TOCOL: `=UNIQUE(TOCOL(A2:C5, 1))`.

Mistake: Seeing repeats in the result, such as "SQL" and "SQL ". **Fix:** Hidden spaces make them different. Clean the data with TRIM.

8. Data Analyst Tip

Add other functions around the formula. `=SORT(UNIQUE(TOCOL(A2:C5, 1)))` gives the list in alphabetical order, and `=COUNTA(UNIQUE(TOCOL(A2:C5, 1)))` counts the different items, which is 4 in the second example.

TOROW Function in Excel

1. What is TOROW?

The TOROW function takes a range with many rows and columns, or a single column, and lays all its values out in a single row. It solves the problem of data that is arranged in a grid or a list when you need it side by side. A Data Analyst uses it to flatten tables and to turn column lists into rows, for example to create column headings for a summary table.

2. Syntax

`=TOROW(array, [ignore], [scan_by_column])`

Argument	Required?	What it accepts
`array`	Required	The range or array to convert into one row (for example `A1:C3`)
`ignore`	Optional	What to skip: `0` keep all (default), `1` ignore blanks, `2` ignore errors, `3` ignore blanks and errors
`scan_by_column`	Optional	`FALSE` (default) reads row by row, left to right. `TRUE` reads column by column, top to bottom.

For a grid with 1, 2, 3 in the first row and 4, 5, 6 in the second: the default gives 1, 2, 3, 4, 5, 6, and `TRUE` gives 1, 4, 2, 5, 3, 6. To use `scan_by_column` and keep the default for `ignore`, leave the middle argument empty: `=TOROW(A1:C3, , TRUE)`.

3. Simple Example

Turn a 3 × 3 grid into one row, ignoring blank cells:

	A	B	C	D	E	F	G
1	Apple	Fig	Lime				
2	Pear		Kiwi				
3	Plum	Date					
4	**Apple**	**Fig**	**Lime**	**Pear**	**Kiwi**	**Plum**	**Date**

Formula in A4 only: `=TOROW(A1:C3, 1)`

Excel reads row 1 (Apple, Fig, Lime), then row 2, then row 3, skipping the two blank cells (B2 and C3). The 7 values spill across A4 to G4. **Result: 7 values in one row**

4. Another Example

Collect the numbers from a grid and remove the errors:

	A	B	C	D
1	120	#N/A	80	
2	95	60	#N/A	
3	**120**	**80**	**95**	**60**

Formula in A3 only: `=TOROW(A1:C2, 2)`

The ignore value 2 tells Excel to skip errors. It reads row 1 (120, 80), then row 2 (95, 60), leaving out both #N/A cells. **Result: 120, 80, 95, 60**. Functions like SUM and AVERAGE can now work on this clean row.

5. When to Use It

- Flatten a multi-column table into one row
- Turn a vertical list into a horizontal row, such as column headings for a summary table
- Combine several columns of values into one row for other formulas
- Remove blanks and errors while reshaping data
- Prepare a row of items for functions such as UNIQUE or SORT

6. Important Things to Remember

- TOROW is a dynamic array function. You type one formula and the results spill to the right.
- By default, TOROW reads row by row. The third argument is `scan_by_column`: FALSE (default) reads by row, TRUE reads by column.
- By default, nothing is ignored, so blank cells appear in the result as 0 and errors are kept.
- Use `1` to skip blanks, `2` to skip errors, and `3` to skip both.
- The cells to the right of the formula must be empty, or Excel shows `#SPILL!`.
- TOROW is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Expecting blank cells to be skipped automatically, then finding 0 values in the result. **Fix:** Add the ignore argument: `=TOROW(A1:C3, 1)`.

Mistake: Using `1` to remove blanks, but still seeing `#N/A` in the result. **Fix:** Use `3` to ignore both blanks and errors, or `2` for errors only.

Mistake: Getting the wrong order because of the reading direction. **Fix:** Row by row is the default. Add `TRUE` as the third argument for column by column.

Mistake: Getting `#SPILL!` because cells to the right of the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

Use TOROW to build headings that update by themselves. For example, `=TOROW(UNIQUE(A2:A20))` lists each unique item from a column across a row. You can use this as the column headings of a summary table, and the headings will grow or shrink with your data.

VSTACK Function in Excel

1. What is VSTACK?

The VSTACK function joins two or more tables or ranges into one, placing each one below the previous one. It solves the problem of having similar data in separate lists, such as sales for different months or employees from different teams. A Data Analyst uses it to consolidate data without copying and pasting.

2. Syntax

`=VSTACK(array1, [array2], ...)`

Argument	Required?	What it accepts
`array1`	Required	The first range or table to stack (for example `A1:B4`)
`array2, ...`	Optional	More ranges or tables to place below the first one, in the order you list them

The result has as many rows as all the arrays combined, and as many columns as the widest array. If you include the header row of the first table only, the result has just one set of headings.

3. Simple Example

Combine two employee lists with the same columns into one:

	A	B	C	D	E
1	Name	Department		**Name**	**Department**
2	Ayesha	Marketing		**Ayesha**	**Marketing**
3	Rahim	Sales		**Rahim**	**Sales**
4	Karim	IT		**Karim**	**IT**
5	Name	Department		**Lina**	**HR**
6	Lina	HR		**Sami**	**Finance**
7	Sami	Finance		**Nusrat**	**Operations**
8	Nusrat	Operations			

Formula in D1 only: `=VSTACK(A1:B4, A6:B8)`

The first list is A1:B4 (with its header). The second list is A6:B8 (data only, without its header in row 5). Excel places the second list below the first. **Result: 1 header row and 6 employees in D1 to E7**

4. Another Example

Combine two stock lists. The second list has no Price column:

	A	B	C	D	E	F	G
1	Item	Qty	Price		**Item**	**Qty**	**Price**
2	Mouse	40	25		**Mouse**	**40**	**25**
3	Keyboard	25	45		**Keyboard**	**25**	**45**
4					**Monitor**	**15**	**#N/A**
5	Item	Qty			**Headset**	**30**	**#N/A**
6	Monitor	15					
7	Headset	30					

Formula in E1 only: `=VSTACK(A1:C3, A6:B7)`

The result is as wide as the widest list (3 columns). The second list has only 2 columns, so Excel fills the missing Price cells with #N/A. To replace them, wrap the formula in IFERROR: `=IFERROR(VSTACK(A1:C3, A6:B7), "")`.

5. When to Use It

- Combine monthly or quarterly sales lists into one table
- Merge lists from different sheets, such as one sheet per region or team
- Build a master list of customers, employees, or products
- Consolidate reports from several departments

6. Important Things to Remember

- VSTACK places the arrays one below the other, in the order you write them.
- VSTACK is a dynamic array function. One formula fills many cells, so the cells where the result spills must be empty, or Excel shows `#SPILL!`.
- If an array has fewer columns than the widest one, the missing cells show `#N/A`.
- Include the header row from the first table only, or the headings will repeat in the middle of the result.
- The result updates when the source data changes, and the sources can be on other sheets.
- VSTACK is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Including the header row of every table, so headings appear again in the middle of the result. **Fix:** Select the header only in the first table.

Mistake: Stacking tables whose columns are in a different order, so values land under the wrong headings. **Fix:** Check that the columns match in the same order before stacking.

Mistake: Seeing `#N/A` because the tables have different numbers of columns. **Fix:** Make the structure match, or wrap the formula in IFERROR.

Mistake: Getting `#SPILL!` because cells below or beside the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

Combine VSTACK with other functions, for example `=SORT(VSTACK(Jan!A2:C20, Feb!A2:C20))` to merge two monthly sheets and sort the result. Select only rows that contain data, because blank rows appear as 0 in the result.

VSTACK + CHOOSECOLS + FILTER in Excel

1. What is VSTACK + CHOOSECOLS + FILTER?

These three functions work together. FILTER keeps only the rows that meet a condition, CHOOSECOLS keeps only the columns you need from those rows, and VSTACK stacks the results from several tables into one list. Together they collect matching records from different tables, such as different regions or months. A Data Analyst uses them to build one clean report from several sources.

2. Syntax

`=VSTACK(CHOOSECOLS(FILTER(array1, include1), col_num1, [col_num2], ...), CHOOSECOLS(FILTER(array2, include2), col_num1, [col_num2], ...))`

Function	Argument	Required?	What it accepts
FILTER	`array`	Required	The range to filter (for example `A2:D4`)
FILTER	`include`	Required	A TRUE/FALSE test for each row, such as `B2:B4="Laptop"`. Join tests with * (AND) or + (OR).
CHOOSECOLS	`col_num1, ...`	col_num1 required	The columns to keep from the filtered rows, counted inside the array. More numbers are optional.
VSTACK	`array1, ...`	array1 required	The CHOOSECOLS results to stack. More arrays are optional.

Excel works from the inside out: FILTER first, then CHOOSECOLS, then VSTACK. Every CHOOSECOLS part must return the same columns in the same order.

3. Simple Example

Collect the laptop sales from the North and East tables:

	A	B	C	D	E	F	G
1	ID	Product	Region	Sales		**Product**	**Sales**
2	P001	Laptop	North	1200		**Laptop**	**1200**
3	P002	Mouse	North	150		**Laptop**	**800**
4	ID	Product	Region	Sales			
5	P004	Laptop	East	800			
6	P005	Mouse	East	250			

Formula in F2 only (headings in F1:G1 are typed):

```
=VSTACK(CHOOSECOLS(FILTER(A2:D3, B2:B3="Laptop"), 2, 4), CHOOSECOLS(FILTER(A5:D6, B5:B6="Laptop"), 2, 4))
```

FILTER keeps the Laptop row in each table. CHOOSECOLS keeps columns 2 (Product) and 4 (Sales). VSTACK stacks the two results.
Result: Laptop 1200 and Laptop 800

4. Another Example

Collect North orders over 300 from the Jan and Feb tables:

	A	B	C	D	E	F
1	Order	Region	Amount		**Order**	**Amount**
2	O101	North	450		**O101**	**450**
3	O102	South	600		**O201**	**520**
4	Order	Region	Amount		**O202**	**310**
5	O201	North	520			
6	O202	North	310			

Formula in E2 only (headings in E1:F1 are typed):

```
=VSTACK(CHOOSECOLS(FILTER(A2:C3, (B2:B3="North")*(C2:C3>300)), 1, 3), CHOOSECOLS(FILTER(A5:C6, (B5:B6="North")*(C5:C6>300)), 1, 3))
```

The * means both conditions must be true. Jan gives O101. Feb gives O201 and O202. **Result: 3 rows**

5. When to Use It

- Collect matching records from several tables, such as one table per region or month
- Compare one product or customer across different tables
- Combine high-value orders from several sources into one list
- Build report lists that show only the columns you need
- Create dynamic lists for dashboards that update when the data changes

6. Important Things to Remember

- Read the formula from the inside out: FILTER, then CHOOSECOLS, then VSTACK.
- Each CHOOSECOLS part must return the same columns in the same order. Fewer columns give `#N/A`.
- If FILTER finds no match in one table, it returns `#CALC!` and the whole formula shows an error. Wrap it in IFERROR to show a message.
- Column numbers in CHOOSECOLS count inside the filtered array. Text in conditions goes in quotes, and each condition goes in brackets.
- The result spills into neighboring cells. They must be empty, or Excel shows `#SPILL!`.
- CHOOSECOLS and VSTACK need Microsoft 365 or Excel 2024. This formula does not work in Excel 2021, even though FILTER does.

7. Common Mistakes

Mistake: One table has no matching rows, so the whole formula shows `#CALC!`. **Fix:** Wrap it in IFERROR, such as `=IFERROR(VSTACK(...), "No matches")`. The `if_empty` argument does not help, because CHOOSECOLS then gives `#VALUE!`.

Mistake: Choosing different columns in each CHOOSECOLS, so `#N/A` appears or values land under the wrong headings. **Fix:** Use the same column numbers in every part.

Mistake: Forgetting the brackets around each condition, such as `B2:B4="North"*C2:C4>300`. **Fix:** Write `(B2:B4="North")*(C2:C4>300)`.

8. Data Analyst Tip

If your data is in Excel Tables, use table names such as `Table1[Region]` in the formula, so it picks up new rows automatically as the tables grow.

WRAPROWS Function in Excel

1. What is WRAPROWS?

The WRAPROWS function takes a single row or column of values and wraps it into a table, filling one row at a time. You decide how many values go in each row. It solves the problem of long lists that are hard to read or print. A Data Analyst uses it to reshape lists into compact tables and to split a list into rows of equal size, such as months into quarters.

2. Syntax

`=WRAPROWS(vector, wrap_count, [pad_with])`

Argument	Required?	What it accepts
`vector`	Required	A single row or a single column of values (for example `A1:A6`). A table with several columns and rows does not work.
`wrap_count`	Required	The maximum number of values in each row, which is the width of the result. A value of 3 means 3 columns.
`pad_with`	Optional	The value to put in any empty cells, such as `""` or `"-"`. The default is `#N/A`.

Excel fills the first row from left to right, then starts a new row below. It works out the number of rows itself: 6 values with a `wrap_count` of 3 make 2 rows, and 7 values make 3 rows with 2 empty cells.

3. Simple Example

Wrap a list of 6 items into rows of 3:

	A	B	C	D	E
1	Apple		**Apple**	**Banana**	**Orange**
2	Banana		**Mango**	**Kiwi**	**Grape**
3	Orange				
4	Mango				
5	Kiwi				
6	Grape				

Formula in C1 only: `=WRAPROWS(A1:A6, 3)`

Each row holds 3 values. Excel fills the first row with Apple, Banana, Orange, then starts a second row with Mango, Kiwi, Grape. **Result: 2 rows and 3 columns**

4. Another Example

Arrange 5 monthly sales figures into rows of 3 (one quarter per row):

	A	B	C	D	E
1	Sales		**Month 1**	**Month 2**	**Month 3**
2	120		**120**	**150**	**130**
3	150		**170**	**160**	**-**
4	130				
5	170				
6	160				

Formula in C2 only: `=WRAPROWS(A2:A6, 3, "-")` (the headings in C1:E1 are typed)

Five values make two rows. Row 1 gets 3 values. Row 2 gets only 2, so the empty cell is filled with the pad value "-" instead of `#N/A`.

5. When to Use It

- Turn a long list into a compact table for reading or printing
- Break daily or monthly figures into weeks or quarters, one row per period
- Split a list into groups of equal size, one group per row
- Arrange list items in a grid layout for a report or dashboard
- Rebuild a table from a flattened list, for example after using TOCOL

6. Important Things to Remember

- WRAPROWS fills row by row: left to right, then down. Its partner WRAPCOLS fills column by column.
- `wrap_count` is the number of values in each row (the table width), not the number of rows.
- `vector` must be a single row or column, or Excel shows `#VALUE!`. A `wrap_count` below 1 gives `#NUM!`.
- Empty cells show `#N/A` unless you give a `pad_with` value.
- WRAPROWS is a dynamic array function. The cells for the result must be empty, or Excel shows `#SPILL!`.
- WRAPROWS is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Thinking `wrap_count` is the number of rows. `=WRAPROWS(A1:A6, 3)` gives 2 rows and 3 columns, not 3 rows. **Fix:** To get 3 rows from 6 values, use a `wrap_count` of 2.

Mistake: Selecting a range with several columns, which gives `#VALUE!`. **Fix:** Use a single row or column. Flatten a table first with TOCOL or TOROW.

Mistake: Leaving `#N/A` in the result because `pad_with` was left out. **Fix:** Add a pad value, such as `""`.

Mistake: Mixing up WRAPROWS and WRAPCOLS and getting the values in the wrong order. **Fix:** WRAPROWS fills across rows. WRAPCOLS fills down columns.

8. Data Analyst Tip

To get a set number of rows, calculate the `wrap_count`. For example, `=WRAPROWS(A2:A20, ROUNDUP(COUNTA(A2:A20)/3, 0), "")` wraps the list into 3 rows, however many items there are.

WRAPCOLS Function in Excel

1. What is WRAPCOLS?

The WRAPCOLS function takes a single row or column of values and wraps it into a table, filling one column at a time. You decide how many values go in each column. It solves the problem of long lists that are hard to read or print. A Data Analyst uses it to reshape lists into compact tables and to split a list into equal groups.

2. Syntax

`=WRAPCOLS(vector, wrap_count, [pad_with])`

Argument	Required?	What it accepts
`vector`	Required	A single row or a single column of values (for example `A1:A6`). A table with several columns and rows does not work.
`wrap_count`	Required	The maximum number of values in each column, which is the height of the result. A value of 3 means 3 rows.
`pad_with`	Optional	The value to put in any empty cells, such as `""` or `"-"`. The default is `#N/A`.

Excel fills the first column from top to bottom, then starts a new column to the right. It works out the number of columns itself: 6 values with a `wrap_count` of 3 make 2 columns, and 7 values make 3 columns with 2 empty cells.

3. Simple Example

Wrap a list of 6 items into a table with 3 rows:

	A	B	C	D
1	Apple		**Apple**	**Mango**
2	Banana		**Banana**	**Kiwi**
3	Orange		**Orange**	**Grape**
4	Mango			
5	Kiwi			
6	Grape			

Formula in C1 only: `=WRAPCOLS(A1:A6, 3)`

Each column holds 3 values. Excel fills the first column with Apple, Banana, Orange, then starts a second column with Mango, Kiwi, Grape. **Result: 3 rows and 2 columns**

4. Another Example

Arrange 5 employees into teams of up to 3, one team per column:

	A	B	C	D
1	Name		**Team 1**	**Team 2**
2	Ayesha		**Ayesha**	**Fahim**
3	Rahim		**Rahim**	**Sadia**
4	Nabila		**Nabila**	**-**
5	Fahim			
6	Sadia			

Formula in C2 only: `=WRAPCOLS(A2:A6, 3, "-")` (the headings in C1:D1 are typed)

Five names make two columns. Team 1 gets 3 names. Team 2 gets only 2, so the empty cell is filled with the pad value "-" instead of `#N/A`.

5. When to Use It

- Turn a long list into a compact table for reading or printing
- Split a list into equal groups, such as teams, with one group per column
- Prepare a compact layout of items for a dashboard
- Reshape a single-column result, such as from UNIQUE, into a table
- Rebuild a table from a flattened list, for example after using TOCOL

6. Important Things to Remember

- WRAPCOLS fills column by column: top to bottom, then left to right. Its partner WRAPROWS fills row by row.
- `wrap_count` is the number of values in each column (the table height), not the number of columns.
- `vector` must be a single row or column, or Excel shows `#VALUE!`. A `wrap_count` below 1 gives `#NUM!`.
- Empty cells show `#N/A` unless you give a `pad_with` value.
- WRAPCOLS is a dynamic array function. The cells for the result must be empty, or Excel shows `#SPILL!`.
- WRAPCOLS is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Thinking `wrap_count` is the number of columns. `=WRAPCOLS(A1:A6, 3)` gives 3 rows and 2 columns, not 3 columns. **Fix:** To get 3 columns from 6 values, use a `wrap_count` of 2.

Mistake: Selecting a range with several columns, which gives `#VALUE!`. **Fix:** Use a single row or column. Flatten a table first with TOCOL or TOROW.

Mistake: Leaving `#N/A` in the result because `pad_with` was left out. **Fix:** Add a pad value, such as `""`.

Mistake: Getting `#SPILL!` because cells where the result will go already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

To get a set number of columns, calculate the `wrap_count`. For example, `=WRAPCOLS(A2:A20, ROUNDUP(COUNTA(A2:A20)/3, 0), "")` wraps the list into 3 columns, however many items there are.

CHOOSECOLS Function in Excel

1. What is CHOOSECOLS?

The CHOOSECOLS function returns only the columns you pick from a table, in the order you choose. It solves the problem of wide tables that have more columns than you need. A Data Analyst uses it to build compact tables, reorder columns, and pull columns that are not next to each other, without copying and pasting.

2. Syntax

`=CHOOSECOLS(array, col_num1, [col_num2], ...)`

Argument	Required?	What it accepts
`array`	Required	The range or table to take columns from (for example `A1:D5`)
`col_num1`	Required	The number of the first column to return. A positive number counts from the left, and a negative number counts from the right (`-1` is the last column).
`col_num2, ...`	Optional	More column numbers. They can be in any order, and the same number can be used more than once.

Column numbers count inside the array, not by the sheet column letters. In `B1:E5`, column number 1 is column B. If you include the heading row in `array`, the headings come along with the data.

3. Simple Example

Return only the Product and Sales columns:

	A	B	C	D	E	F
1	Product	Category	Sales		**Product**	**Sales**
2	Laptop	Electronics	1200		**Laptop**	**1200**
3	Mouse	Electronics	650		**Mouse**	**650**
4	Chair	Furniture	300		**Chair**	**300**
5	Desk	Furniture	750		**Desk**	**750**

Formula in E1 only: `=CHOOSECOLS(A1:C5, 1, 3)`

Excel takes column 1 (Product) and column 3 (Sales) from A1:C5 and skips column 2 (Category). The headings in row 1 come along because row 1 is in the array. **Result: 2 columns and 5 rows**

4. Another Example

Return Salary first, then Name, counting Salary from the right:

	A	B	C	D	E	F	G
1	ID	Name	Dept	Salary		**Salary**	**Name**
2	E101	Ayesha	IT	50000		**50000**	**Ayesha**
3	E102	Rahim	HR	42000		**42000**	**Rahim**
4	E103	Nabila	IT	61000		**61000**	**Nabila**
5	E104	Tanvir	Sales	38000		**38000**	**Tanvir**

Formula in F1 only: `=CHOOSECOLS(A1:D5, -1, 2)`

The -1 means the last column of the array (Salary). The 2 means the second column (Name). Excel returns the columns in the order you list them, so Salary comes before Name.

5. When to Use It

- Pull only the columns you need from a wide dataset for a report or dashboard
- Reorder the columns of a table without moving the source data
- Return columns that are not next to each other, such as the first and fourth
- Always return the last column of a table, even when columns are added (-1)
- Select columns from the result of another function, such as FILTER or SORT

6. Important Things to Remember

- CHOOSECOLS is a dynamic array function. One formula fills many cells, so the cells for the result must be empty, or Excel shows `#SPILL!`.
- Positive column numbers count from the left. Negative column numbers count from the right.
- Columns appear in the order you list them. Repeating a column number repeats that column.
- A column number of 0, or a number larger than the number of columns, gives `#VALUE!`.
- CHOOSECOLS returns values only, not the source formatting. Format the result separately.
- CHOOSECOLS is available in Microsoft 365 and Excel 2024. In Excel 2021 or earlier it shows `#NAME?`.

7. Common Mistakes

Mistake: Using the sheet column number instead of the position inside the array, such as 4 for column D in `B1:D5`, which gives `#VALUE!`. **Fix:** Count from the first column of the array. In `B1:D5`, column D is number 3.

Mistake: Using 0 as a column number. **Fix:** Use 1 or more to count from the left, or -1 or less to count from the right.

Mistake: Getting `#SPILL!` because cells where the result will go already contain data. **Fix:** Clear those cells or move the formula to an empty area.

Mistake: Expecting the number formats and colors of the source to carry over. **Fix:** Apply the formatting to the result range yourself.

8. Data Analyst Tip

Combine CHOOSECOLS with other functions. For example, `=CHOOSECOLS(FILTER(A2:D5, C2:C5="IT"), 2, 4)` returns only the Name and Salary of the IT employees: Ayesha 50000 and Nabila 61000. This gives a compact report in one formula.

CHOOSECOLS + VSTACK in Excel

1. What is CHOOSECOLS + VSTACK?

CHOOSECOLS and VSTACK are two functions used together. CHOOSECOLS picks the columns you want from a table, in the order you want them, and VSTACK stacks tables one below the other. Together they solve the problem of combining tables that have extra columns or a different column order. A Data Analyst uses them to merge data from several sources into one clean table.

2. Syntax

`=VSTACK(CHOOSECOLS(array1, col_num1, [col_num2], ...), CHOOSECOLS(array2, col_num1, [col_num2], ...))`

Function	Argument	Required?	What it accepts
CHOOSECOLS	`array`	Required	The range to take columns from (for example `A2:C3`)
CHOOSECOLS	`col_num1`	Required	The number of the first column to return. A negative number counts from the right.
CHOOSECOLS	`col_num2, ...`	Optional	More column numbers, in the order you want them
VSTACK	`array1`	Required	The first range, or CHOOSECOLS result, to stack
VSTACK	`array2, ...`	Optional	More ranges or results, placed below the first

Excel works from the inside out: CHOOSECOLS selects the columns first, then VSTACK stacks the results.

3. Simple Example

Stack two sales tables, keeping only the ID and Product columns:

	A	B	C	D	E	F
1	ID	Product	Sales		**ID**	**Product**
2	P001	Laptop	1200		**P001**	**Laptop**
3	P002	Mouse	150		**P002**	**Mouse**
4					**P004**	**Monitor**
5	ID	Product	Sales		**P005**	**Headphones**
6	P004	Monitor	800			
7	P005	Headphones	250			

Formula in E2 only (headings in E1:F1 are typed):
`=VSTACK(CHOOSECOLS(A2:C3, 1, 2), CHOOSECOLS(A6:C7, 1, 2))`
The first table is in A1:C3 and the second in A5:C7. CHOOSECOLS keeps columns 1 and 2 of each and drops Sales. VSTACK puts the second result below the first. **Result: 4 rows and 2 columns**

4. Another Example

Stack online and store orders. The store table has Amount and Customer in a different order:

	A	B	C	D	E	F	G
1	Order	Customer	Amount		**Order**	**Customer**	**Amount**
2	O101	Ayesha	250		**O101**	**Ayesha**	**250**
3	O102	Rahim	120		**O102**	**Rahim**	**120**
4					**O201**	**Nabila**	**80**
5	Order	Amount	Customer		**O202**	**Tanvir**	**300**
6	O201	80	Nabila				
7	O202	300	Tanvir				

Formula in E2 only (headings in E1:G1 are typed): `=VSTACK(A2:C3, CHOOSECOLS(A6:C7, 1, 3, 2))`
The online orders (A2:C3) are already in the right order. For the store orders, CHOOSECOLS returns columns 1, 3, 2, which puts Customer before Amount. Now both parts match, and VSTACK stacks them.

5. When to Use It

- Combine monthly or regional tables, keeping only the columns you need
- Merge tables from different sources where the column order is different
- Build a master list of customers, orders, or products from several tables
- Prepare a compact, consistent dataset for a report or dashboard

6. Important Things to Remember

- Read the formula from the inside out: CHOOSECOLS selects the columns, then VSTACK stacks the results.
- Every part should have the same columns in the same order. If one has fewer columns, the missing cells show `#N/A`.
- Column numbers count inside each array. A negative number counts from the right. A 0 or a number that is too large gives `#VALUE!`.
- Include the heading row from the first table only, or type the headings yourself.
- The result spills into neighboring cells. They must be empty, or Excel shows `#SPILL!`.
- Both functions are available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Selecting a different number of columns from each table, so `#N/A` appears. **Fix:** Choose the same number of columns from every table.

Mistake: Stacking tables with different column orders, so values land under the wrong headings. **Fix:** Use CHOOSECOLS to put the columns in the same order first.

Mistake: Including the heading row of the second table, so the headings repeat in the middle. **Fix:** Start each later range at its first data row.

8. Data Analyst Tip

Wrap the formula in another function. For example, `=SORT(VSTACK(A2:C3, CHOOSECOLS(A6:C7, 1, 3, 2)), 3, -1)` stacks the orders and sorts them by Amount, highest first.

CHOOSECOLS + XLOOKUP in Excel

1. What is CHOOSECOLS + XLOOKUP?

XLOOKUP finds a value in a list and returns data from the matching row. When it returns several columns at once, CHOOSECOLS can keep only the columns you need. Together they solve the problem of looking up one record and showing just a few of its fields, such as a name and a salary. A Data Analyst uses them to build compact lookup results without writing a separate lookup for every column.

2. Syntax

`=CHOOSECOLS(XLOOKUP(lookup_value, lookup_array, return_array), col_num1, [col_num2], ...)`

Function	Argument	Required?	What it accepts
XLOOKUP	`lookup_value`	Required	The value to find, such as a cell reference or text in quotes
XLOOKUP	`lookup_array`	Required	The single column to search (for example `A2:A6`)
XLOOKUP	`return_array`	Required	The columns to return from the matching row (for example `B2:D6`). It needs the same number of rows as `lookup_array`.
CHOOSECOLS	`col_num1`	Required	The number of the first column to keep, counted inside `return_array`. A negative number counts from the right.
CHOOSECOLS	`col_num2, ...`	Optional	More column numbers, in the order you want them

Because `return_array` has several columns, XLOOKUP returns the whole matching row. CHOOSECOLS then keeps only the columns you list. XLOOKUP finds an exact match by default.

3. Simple Example

Look up product P004 and return its Name and Sales:

	A	B	C	D	E	F	G	H
1	ID	Name	Region	Sales		Find	**Name**	**Sales**
2	P001	Laptop	North	1200		P004	**Monitor**	**800**
3	P002	Mouse	South	150				
4	P003	Keyboard	North	300				
5	P004	Monitor	East	800				
6	P005	Webcam	South	180				

Formula in G2 only: `=CHOOSECOLS(XLOOKUP(F2, A2:A6, B2:D6), 1, 3)`

XLOOKUP finds P004 and returns its row from B2:D6: Monitor, East, 800. CHOOSECOLS keeps columns 1 and 3 of that row. **Result: Monitor and 800**

4. Another Example

Look up two employee IDs. One of them does not exist:

	A	B	C	D	E	F	G	H
1	ID	Name	Dept	Salary		Find	**Name**	**Salary**
2	E101	Ayesha	IT	50000		E103	**Nabila**	**61000**
3	E102	Rahim	HR	42000		E105	**Not Found**	
4	E103	Nabila	IT	61000				
5	E104	Tanvir	Sales	38000				

Formula in G2, copied to G3:
`=IFERROR(CHOOSECOLS(XLOOKUP(F2, A2:A5, B2:D5), 1, 3), "Not Found")`

E103 is found, so you get its Name and Salary. E105 is not in the list, so XLOOKUP gives an error and IFERROR shows "Not Found" instead of `#N/A`.

5. When to Use It

- Look up a record by ID and show only some of its fields
- Build lookup cards or dashboards that show a few fields for a selected item
- Return several fields with one formula instead of one lookup per column
- Return columns that are not next to each other from a matching row
- Avoid copying columns you do not need into a report

6. Important Things to Remember

- XLOOKUP finds the row. CHOOSECOLS picks the columns from that row.
- Column numbers in CHOOSECOLS count inside `return_array`, not by sheet column letters. In `B2:D6`, column 1 is B.
- `lookup_array` must be a single column, and `return_array` must have the same number of rows.
- The result spills across columns. The cells to the right must be empty, or Excel shows `#SPILL!`.
- If the lookup value is not found, the result is `#N/A`. Wrap the whole formula in IFERROR to show a message.
- CHOOSECOLS is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier, even though XLOOKUP is.

7. Common Mistakes

Mistake: Using the sheet column number in CHOOSECOLS, such as 4 for column D in `B2:D6`, which gives `#VALUE!`. **Fix:** Count from the first column of `return_array`.

Mistake: Using ranges with different numbers of rows, such as `A2:A6` with `B2:D7`. **Fix:** Select the same rows for both.

Mistake: Using the `if_not_found` argument inside XLOOKUP. It returns one value, and CHOOSECOLS then gives `#VALUE!`. **Fix:** Put IFERROR around the whole formula.

Mistake: Selecting several columns as `lookup_array`, such as `A2:B6`. **Fix:** Search in a single column.

8. Data Analyst Tip

You can also put CHOOSECOLS inside XLOOKUP: `=XLOOKUP(F2, A2:A6, CHOOSECOLS(B2:D6, 1, 3))` gives the same result. Use the version you find easier to read. Put the lookup value in a cell and lock the ranges with $, such as `$A$2:$A$6`, so you can copy the formula down.

IFERROR Function in Excel

1. What is IFERROR?

The IFERROR function checks a formula. If the formula gives an error, IFERROR shows a value you choose instead. If there is no error, it shows the normal result. It solves the problem of messy error messages such as #DIV/0! appearing in reports. A Data Analyst uses it to keep tables and dashboards clean and easy to read.

2. Syntax

`=IFERROR(value, value_if_error)`

Argument	Required?	What it accepts
`value`	Required	The formula, expression, or cell to check for an error (for example `A2/B2`)
`value_if_error`	Required	What to show if there is an error: text in quotes (`"Error"`), a number (`0`), empty quotes (`""`) for a blank look, or a cell reference

Both arguments are required. IFERROR does not change the result when there is no error; it only steps in when the first argument returns an error.

3. Simple Example

Divide Number by Divide By, and show "Error" if the division fails:

	A	B	C
1	Number	Divide By	**Result**
2	10	2	**5**
3	10	0	**Error**
4	15	5	**3**
5	12	0	**Error**

Formula in C2: `=IFERROR(A2/B2, "Error")`, then copied down to C5.

Row 2: 10 ÷ 2 works, so Excel shows 5. Row 3: 10 ÷ 0 would give #DIV/0!, so Excel shows "Error" instead. Rows 4 and 5 follow the same rule.

4. Another Example

Find the average sales per customer for each region:

	A	B	C	D
1	Region	Sales	Customers	**Per Customer**
2	North	50000	10	**5000**
3	South	24000	0	**No data**
4	East	36000	12	**3000**
5	West	0	0	**No data**

Formula in D2: `=IFERROR(B2/C2, "No data")`, then copied down to D5.

North works normally: 50000 ÷ 10 = 5000. South has 0 customers, and West gives 0 ÷ 0. Both would show #DIV/0!, so Excel shows "No data" instead.

5. When to Use It

- Avoid #DIV/0! in ratios, percentages, and averages per customer or per order
- Replace #N/A in lookup formulas with a clear message such as "Not found"
- Keep reports and dashboards clean before sharing them with managers
- Replace errors with a blank or a custom value during data cleaning
- Prevent errors from spreading into other formulas that depend on the result

6. Important Things to Remember

- IFERROR catches every type of error: `#DIV/0!`, `#N/A`, `#VALUE!`, `#REF!`, `#NAME?`, `#NUM!`, and `#NULL!`.
- If the formula has no error, IFERROR returns its normal result unchanged.
- Text replacements go in quotes (`"No data"`). Numbers do not. Empty quotes (`""`) make the cell look blank.
- Choose the replacement carefully. A 0 is counted in totals and averages, while text and `""` in a range are ignored by SUM and AVERAGE.
- Because IFERROR hides all errors, it can also hide real mistakes, such as a misspelled function name or a wrong cell reference.
- Both arguments are required.

7. Common Mistakes

Mistake: Adding IFERROR too early, which hides real mistakes in the formula. **Fix:** Build and test the formula first, then add IFERROR at the end.

Mistake: Forgetting quotation marks around a text replacement, such as `=IFERROR(A2/B2, Error)`. This causes `#NAME?`. **Fix:** Write `"Error"` in quotes.

Mistake: Using 0 as the replacement when 0 would distort the results, such as in an average. **Fix:** Use `""` or a text message like `"No data"` when the value should not count as zero.

Mistake: Leaving out the second argument, such as `=IFERROR(A2/B2)`. **Fix:** Always give both arguments: the value and the replacement.

8. Data Analyst Tip

Use a clear message such as "No data" instead of 0 when a division fails, because 0 looks like a real result. Add IFERROR as the last step, after checking that the formula works on normal rows. This way you only hide the errors you expect, and real mistakes still get noticed.

TRIM + CLEAN in Excel

1. What are TRIM and CLEAN?

TRIM removes unnecessary spaces from text: spaces at the start, spaces at the end, and repeated spaces between words. CLEAN removes non-printable characters, such as hidden line breaks and tabs, that you cannot see in the cell. Both solve the problem of messy text that looks fine but does not match when you count, look up, or group it. A Data Analyst uses them together as a first step when cleaning data imported from websites, files, or databases.

2. Syntax

Function	Syntax	What it removes
TRIM	`=TRIM(text)`	Leading and trailing spaces, and extra spaces between words
CLEAN	`=CLEAN(text)`	Non-printable characters, such as line breaks and tabs
Together	`=TRIM(CLEAN(text))`	Both: CLEAN runs first, then TRIM

Each function has one required argument, `text`: a cell reference (for example `A2`) or text typed in quotes. There are no optional arguments.

3. Simple Example

Clean names from an imported file. Quotation marks show the spaces, and ¶ marks a hidden line break:

	A	B
1	Raw Name	**Clean Name**
2	" Abu Nousher "	**"Abu Nousher"**
3	"Rahim Khan¶"	**"Rahim Khan"**
4	"Tanvir Ahmed¶ "	**"Tanvir Ahmed"**

Formula in B2: `=TRIM(CLEAN(A2))`, then copied down to B4.

CLEAN first removes the hidden line breaks in rows 3 and 4. TRIM then removes the extra spaces. **Result: clean names with single spaces and no hidden characters**

4. Another Example

Clean addresses where a line break (¶) sits between two words:

	A	B	C
1	Address	CLEAN only	**Fixed**
2	12 Road 5¶Dhaka	12 Road 5Dhaka	**12 Road 5 Dhaka**
3	House 8¶Sylhet	House 8Sylhet	**House 8 Sylhet**

Formula in C2: `=TRIM(CLEAN(SUBSTITUTE(A2, CHAR(10), " ")))`

CLEAN deletes a line break without adding a space, so the words join together (column B). SUBSTITUTE first replaces the line break, CHAR(10), with a space, so the words stay apart (column C).

5. When to Use It

- Clean names, addresses, product codes, and email addresses imported from websites, CSV files, or databases
- Fix lookups, counts, and filters that fail because of hidden spaces or characters
- Prepare a column before using UNIQUE, a pivot table, or a lookup
- Clean text copied from PDFs or web pages
- Standardize text before combining or comparing columns

6. Important Things to Remember

- TRIM keeps one single space between words. It never removes the space between two words completely.
- CLEAN removes characters with codes 0 to 31, such as line breaks and tabs. It removes them without adding a space.
- Neither function removes non-breaking spaces (common in web data). Replace them first with `SUBSTITUTE(A2, CHAR(160), " ")`.
- Write `TRIM(CLEAN(A2))` with CLEAN inside. CLEAN runs first, then TRIM removes any spaces that are left.
- The result is text, and the original cell stays unchanged. Put the formula in a new column.
- Both functions work on one cell at a time. Copy the formula down the column.

7. Common Mistakes

Mistake: Using CLEAN when a line break separates two words, so the words join ("Road 5Dhaka"). **Fix:** Replace the line break with a space first: `SUBSTITUTE(A2, CHAR(10), " ")`.

Mistake: Thinking the formula failed because the text looks unchanged. **Fix:** The data may contain non-breaking spaces. Use `SUBSTITUTE(A2, CHAR(160), " ")` before TRIM.

Mistake: Deleting the original column while the formulas still depend on it, which gives `#REF!`. **Fix:** Copy the cleaned results and paste them as values first.

Mistake: Writing `CLEAN(TRIM(A2))`. **Fix:** Use `TRIM(CLEAN(A2))` so TRIM can remove spaces left behind by CLEAN.

8. Data Analyst Tip

Check which cells need cleaning before you change anything. Type `=LEN(A2)-LEN(TRIM(CLEAN(A2)))` in a helper column. A result above 0 shows how many unwanted characters are in that cell, so you can filter for them and clean only what is needed.

AGGREGATE Function in Excel

1. What is AGGREGATE?

The AGGREGATE function calculates a summary of a list, such as a sum, average, count, maximum, median, or k-th largest value, and lets you choose what to ignore: hidden rows, error values, or both. It solves the problem that SUM and AVERAGE fail when one cell has an error, and SUBTOTAL cannot skip errors. A Data Analyst uses it to get clean results from messy data without helper columns.

2. Syntax

Reference form: `=AGGREGATE(function_num, options, ref1, [ref2], ...)`
Array form: `=AGGREGATE(function_num, options, array, [k])`

Argument	Required?	What it accepts
`function_num`	Required	A number from 1 to 19 that chooses the calculation (see the first table below)
`options`	Required	A number from 0 to 7 that chooses what to ignore (see the second table below)
`ref1 / array`	Required	A range, cell reference, or array to calculate (for example `B2:B6`)
`ref2, ... / k`	Optional	Reference form: more ranges, up to 253. Array form: k, which functions 14 to 19 need.

#	Function	#	Function	#	Function	#	Function	#	Function
1	AVERAGE	5	MIN	9	SUM	13	MODE.SNGL	17	QUARTILE.INC
2	COUNT	6	PRODUCT	10	VAR.S	14	LARGE	18	PERCENTILE.EXC
3	COUNTA	7	STDEV.S	11	VAR.P	15	SMALL	19	QUARTILE.EXC
4	MAX	8	STDEV.P	12	MEDIAN	16	PERCENTILE.INC		

Option	0	1	2	3	4	5	6	7
Ignores hidden rows	No	Yes	No	Yes	No	Yes	No	Yes
Ignores error values	No	No	Yes	Yes	No	No	Yes	Yes
Ignores nested SUBTOTAL / AGGREGATE	Yes	Yes	Yes	Yes	No	No	No	No

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	70
3	Karim	80
4	Hasan	#N/A
5	Nabil	90
6	**Total**	**240**

Formula: `=AGGREGATE(9, 6, B2:B5)`
Function 9 is SUM and option 6 ignores errors. Excel skips #N/A and adds 70 + 80 + 90. **Result: 240**. A plain SUM would return #N/A.

4. Another Example

	A	B
1	Product	Monthly Sales
3	Mouse	3500
4	Keyboard	4500
5	Monitor	8000
6	Headset	6000
7	**2nd Largest**	**6000**

Formula: `=AGGREGATE(14, 5, B2:B6, 2)`
Function 14 is LARGE, option 5 ignores hidden rows, and k = 2. A filter hides row 2 (Laptop, 12000), so the 2nd largest visible value is 6000. **Result: 6000**. A plain LARGE would give 8000.

5. When to Use It

- Sum or average a column that contains errors such as #N/A or #DIV/0!
- Find the largest, smallest, or k-th value among the visible rows only
- Get the median, percentile, or quartile of filtered data
- Count the numbers in a range while skipping errors and hidden rows

6. Important Things to Remember

- Functions 1 to 13 use the reference form. Functions 14 to 19 (LARGE, SMALL, PERCENTILE, QUARTILE) need k.
- Hidden rows means filtered rows and rows hidden by hand. Hidden columns are never ignored.
- Options 0 to 3 ignore nested SUBTOTAL and AGGREGATE formulas. Options 4 to 7 count them. Empty options works like 0.
- If the option you choose does not ignore an error, the result shows that error. An invalid function_num or option, a missing k, or a 3-D reference returns `#VALUE!`.

7. Common Mistakes

Mistake: Leaving options empty or typing 0 and expecting errors to be skipped. **Fix:** Use 6 to ignore errors, or 7 for hidden rows and errors.

Mistake: Using function 14 to 19 without k, such as `=AGGREGATE(14, 6, B2:B6)`. **Fix:** Add k: `=AGGREGATE(14, 6, B2:B6, 2)`.

Mistake: Expecting hidden columns to be ignored. No option skips them. **Fix:** Filter rows instead.

Mistake: Skipping errors without asking why they exist. A skipped error can hide bad data. **Fix:** Resolve the source when you can, and count what was skipped (see the tip).

8. Data Analyst Tip

Always compare the result with the data it came from. Next to it, calculate `=AGGREGATE(2, 7, B2:B6)` to see how many numbers were used, and compare it with `=ROWS(B2:B6)`. If it is lower, some cells are errors, text, or hidden, so the result covers part of the data.

SUBTOTAL Function in Excel

1. What is SUBTOTAL?

The SUBTOTAL function calculates a summary of a range, such as a sum, average, count, maximum, or minimum, and lets you control how hidden and filtered rows are treated. It solves the problem that SUM and AVERAGE still count rows you have filtered out. A Data Analyst uses it to get totals that match exactly what is visible on screen. Excel's Data > Subtotal command and a Table's Total Row use it behind the scenes.

2. Syntax

`=SUBTOTAL(function_num, ref1, [ref2], ...)`

Argument	Required?	What it accepts
`function_num`	Required	A number that chooses the calculation: 1 to 11, or 101 to 111 (see the table below)
`ref1`	Required	A range or cell reference to calculate (for example `B2:B5`)
`ref2, ...`	Optional	More ranges or references. Excel accepts up to 254 of them.

Calculation	SUM	AVERAGE	COUNT	COUNTA	MAX	MIN
function_num (includes manually hidden rows)	9	1	2	3	4	5
function_num (ignores manually hidden rows)	109	101	102	103	104	105

Most often you will use `=SUBTOTAL(9, B2:B5)` for a total, or `=SUBTOTAL(109, B2:B5)` to also skip rows you hid by hand.

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	70
3	Karim	80
4	Hasan	60
5	Nabil	90
6	**Subtotal**	**300**

Formula: `=SUBTOTAL(9, B2:B5)`
Excel adds 70 + 80 + 60 + 90 and returns **Result: 300**. With no hidden rows, this matches SUM.

4. Another Example

	A	B
1	Product	Monthly Sales
2	Laptop	12000
3	Mouse	3500
5	Monitor	8000
6	Headset	6000
7	**Subtotal**	**29500**

Formula: `=SUBTOTAL(9, B2:B6)`
A filter hides row 4 (Keyboard, 4500). SUBTOTAL skips it: 12000 + 3500 + 8000 + 6000 = 29500. **Result: 29500**. A plain SUM would still give 34000.

5. When to Use It

- Total only the visible rows of a filtered sales list
- Count how many rows a filter is currently showing
- Find the average, maximum, or minimum of the filtered rows
- Add group subtotals to a list without double counting them in the grand total
- Build a report where you hide rows by hand and the total must update

6. Important Things to Remember

- SUBTOTAL applies the function you choose (SUM, AVERAGE, COUNT and so on) only to the rows that count.
- Codes 1 to 11 include rows you hid by hand. Codes 101 to 111 ignore them. Both always ignore rows hidden by a filter.
- Any other SUBTOTAL formulas inside the range are ignored, so a grand total never double counts.
- Hidden columns are not ignored by either set of codes. SUBTOTAL is built for rows of data in a column.
- If a referenced cell contains an error such as #N/A or #DIV/0!, the SUBTOTAL result shows that error.
- A function_num that is not valid (for example 12) or a 3-D reference returns `#VALUE!`.

7. Common Mistakes

Mistake: Using SUM on a filtered list. SUM still adds the rows the filter has hidden. **Fix:** Use SUBTOTAL so the total matches the visible rows.

Mistake: Mixing up 9 and 109. Code 9 still counts rows you hid by hand. **Fix:** Use 109 (or 101 to 111 for other functions) when hidden rows must not count.

Mistake: Expecting hidden columns to be ignored. Neither set of codes skips them. **Fix:** Filter rows instead, or point the formula only at the columns you want.

Mistake: Including the Subtotal cell itself, or an old Total row, in the range. **Fix:** Select only the data rows. Including the formula cell causes a circular reference.

8. Data Analyst Tip

Always compare the SUBTOTAL with the data it came from. Next to your total, calculate `=SUBTOTAL(103, A2:A6)` to see how many rows are visible, and compare it with `=COUNTA(A2:A6)` for all rows. If the visible count is lower, some rows are hidden, so your total covers only part of the dataset.